Identifying Root Causes of Drug Addiction Using Explainable AI (XAI)

Final Year Design Project

By

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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements for the **Degree of Bachelor of Science in Computer Science and Engineering**

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APPROVAL

This Project titled "**Identifying Root Causes of Drug Addiction Using Explainable AI (XAI)**" submitted by [**Shoaib Hossain**] to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **06-01-2026**.

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We hereby declare that this project has been done by us under the supervision of **Mr. Md. Shah Jalal**, **Senior Lecturer**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

Drug abuse remains a major public health concern in Bangladesh, particularly among young populations, and is influenced by multiple socio-economic and psycho-social factors. This study integrates machine learning with Explainable Artificial Intelligence (XAI) to identify and interpret the root causes of drug addiction. Data were collected from rehabilitation patients with approval from the Department of Narcotics Control (DNC) and from diverse individuals through online sources. After rigorous preprocessing, predictive models were trained and interpreted using SHAP and LIME.Grouped SHAP analysis shows that socio-economic factors have the highest influence on addiction risk (total SHAP impact = 1.83), followed by psycho-social factors (1.70). Environmental (0.88) and other factors (0.79) play comparatively smaller roles. The findings highlight the importance of age, early drug exposure, family relations, psychological stress, and peer influence. Overall, the study demonstrates that XAI provides transparent and reliable explanations, supporting data-driven policy making and targeted intervention strategies for addressing drug addiction in Bangladesh.

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Chapter 1

Introduction

This chapter summarizes the project's context, problem background, motivation for using XAI, and briefly outlines the objectives, methodology, expected outcomes, and report structure.

1.1 Introduction

Addiction to drugs is a chronic, relapsing brain disease that is characterized by compulsive drug seeking and use, despite its devastating health consequences. It has evolved into a significant global public health issue, with the United Nations Office on Drugs and Crime (UNODC) estimating that some 316 million people - nearly 6% of the world's population aged 15–64 years - used illicit substances in 2023 [1],[2]. The sharp incline in substance abuse emphasizes the immediate requirement for evidence-based methodologies that can reliably pinpoint the underlying risk factors and offer solutions for its prevention.

And in Bangladesh, the spike in drug addiction is also being compared with a runaway train—by 2025, according to estimates by the Department of Narcotics Control (DNC), as many as nearly 8.3 million people will be turned into drug addicts — or 4.89% of the country's entire population! Most of these users are 15–35 years old, signifying an emerging youth epidemic. The most widely abused substances are cannabis, Yaba, alcohol, opioids and sedative showing easy access and widespread misuse among various socio-economic groups [3],[5]. The causes of drug addiction in Bangladesh are multi-factorial including contradictions with family, unemployment, friends and peer pressure, mental health problems, poor institutional support as well. These influences differ from person to person and are the reasons that addiction is a complex problem, one that has not been explained by traditional methods[9].

Explainable AI (XAI) is one of the most promising approaches to address the challenge by offering verifiable explanations behind model predictions. Whereas black box machine learning provides the prediction, XAI methodologies including SHAP and LIME serve to demystify which factors matter for an individual's risk of addiction so that stakeholders can comprehend both the pattern and type of reasoning reflected in the analysis.

This work applies XAI technique to predict the reasons for the drug addiction of Bangladesh by coupling predictive modelingand transparent interpretations together. The aim is to produce policy-relevant, evidence-based knowledge for early detection and effective interventions for Bangladesh's social-cultural setting.

1.2 Motivation

Drug dependence is a complicated and multivariable disorder determined by social, psychological, economic and environmental circumstances. Conventional methodologies have difficulty uncovering the underlying interactions between these various components that can potentially shed a light on the true causes of addiction. This serves as a great incentive to apply new computational approaches that are able to tackle large quantities of data and extract relevant patterns which regular methods sometimes fail to recognize.

On the technical side of things, today's machine learning methods boast impressive predictive performances, but most models also act as "black boxes", spewing out predictions without revealing how they arrived at them. In a sensitive area such as drug addiction, obscurity is significant shortcoming since health care providers and policymakers need to know exactly why/what they are predicting before implementing an intervention. This challenge is tackled by Explainable Artificial Intelligence (XAI), an initiative that focuses on the interpretability of model decisions with a mission to make AI-generated insights trustworthy for end-users while guaranteeing fairness, transparency, and ethical use of data-driven systems.

There are also significant academic and personal rewards to this study. In progressing this project we develop expertise in techniques for data science, machine learning and explainability, investigative methodology and ethical AI development – all of which are crucial skills for modern computer sciences. Finally, also of great value is the development of our in-demand skills as data engineers on problems that have social impact (which improves career readiness and future job prospects working with AI, healthcare analytics, and public sector tech).

1.3 Objectives

- ➢ Determine main socio-economic, psychosocial and environmental determinants of drug-addiction with an in-depth data analysis.
- ➢ Leverage techniques from Explainable AI (XAI) to provide transparent, interpretable and trustworthy model decisions.
- ➢ Create evidence-based knowledge and practices for preventing and mitigating the risk of early drug addiction.

1.4 Methodology

This study design is quantitative, cross-sectional. A structured questionnaire was prepared and pretested and hand-distributed to different institutions for pneumatic prophylaxis in order to ensure authenticity and a wide variety of answers. The methodology includes:

- Development and validation of survey tool for socio-economic, psycho-social and environmental variables.
- A literature review on drug addiction and Explainable AI (XAI).
- Final research objectives, the method of data collection (interviews and survey), and an XAI-based analysis plan were reached.
- The data were gathered through face-to-face interviews (rehabilitation center and DNC referral) and online surveys (Google Forms).
- Loading, cleaning and prepping data with spreadsheets & Python
- Statistical analysis Descriptive and inferential analysis.
- Constructing features related to behavioral, psychological, and environmental risk determinants in the engineering work.
- Development and validation of predictive machine learning model for drug addiction analysis.
- Interpret model results employing Explainable AI (SHAP, LIME, and permutation importance) to pinpoint root cause effects.

1.5 Project Outcome

Expected outcomes of the project include:

- A pre-processed, well-characterized dataset on factor affecting of drug addiction in bangladesh.
- Evidence based statistics and visualized data about SES, PSY, and ENV risk factor.
- Predictive machine learning models for patterns and risk of drug addiction.
- KHF-I with transparent XAI-based interpretability results for the most important contributing factors.
- Evidence-based information for drug prevention and intervention strategies
- A full and transparent (methods) framework that can be used for future research and policy-level usages.

1.6 Organization of the Report

The paper is structured as follows:

Chapter 1 – Introduction: Background and motivation to the study: Introduction drug addiction related research context; Rationale of the current workAims of the study: Campaign of aims and expected outcomesAnticipated findings/Possible conclusion/scientific merits Analysis Methodology to be implied also present.;

Chapter 2 – Literature review and Background: It reviews the literature on drug dependence; socio-economic and psycho-social correlates of it, use of XAI as well as gap in research covered in this study.

3.Methodology: Describes the research design, data collection process, data preprocessing techniques, feature selection in combination with machine learning model and its interpretability that are inspired by XAI.

4 Implementation and Results: Describes the properties, exploratory data analysis of dataset, summary statistics results, model-performance comparison and interpretation of characteristics in root causes based on XAI techniques.

Chapter 5 – Standards and Engineering Impact: Covers engineering best practice, ethical and privacy considerations as well as the potential wider societal impact of the research and typical project management problems.

6 Conclusions and Future Work: Summarizes the findings, describes limitations of this study, and suggests future research and system development.

Chapter 2

Background

This chapter presents the key concepts, reviews related work, discusses similar applications, identifies research gaps, and provides a brief summary.

2.1 Introduction

Drug addiction in Bangladesh arises due to individual, social and environmental issues precipitated by poverty, unemployment, illiteracy and inadequate mental health services [4], [5]. Approximately 80% of the users are aged 16–35 and frequently have financial dependence on family, thereby increasing economic stress [5]. Dysfunctional family, unemployment and peer pressure still the main factors that point to the concerned education and social support resort along with enforcement action [9].

ML models, e.g. Random Forest (RF), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) achieve an accuracy over in predicting high risk cases, while XAI tools, e.g. LIME explicate how stress, impulsiveness, and personality affect drugs [10]. Responsible AI encourages transparency, fairness, and privacy preservation to avoid biases and sexual stigma [11].

Internationally, research in UK has identified living independently of the parents, religiosity or practice and nightlife behaviors as the factors which encourage drug use amongst youth [12]. Interpretability of XAI fosters confidence and supports adoption in health policy, clinical care, and social programs. This research uses XAI to diagnose the root causes of drug addiction in Bangladesh, and provides transparent and interpretable insights for prevention, treatment as well as regulations.

2.2 Literature Review

Drug addiction research in the past has looked at social, family and lifestyle factors that associate with drug use[9]. Identified sociocultural and family related factors influencing drug addiction in Bangladesh[6]. whereas found that nightlife attendance and living apart increase drug use among students. While informative, these early analyses were primarily based on questionnaires and statistics which could provide little predictive power.

Thanks to the progress achieved in machine learning, various investigation investigated computational potential for predicting CSB risk. [7],[5]. also obtained high accuracy (97.9% and 95%) through ML models, yet the methods were not interpretable. To address this issue, [5] used XAI techniques including LIME, which explained how stress, impulsivity and personality traits affect drug use decisions.

Predictive models were also used in related studies to understand smoking- and alcohol-attributable risks[13],[15].combined Logistic Regression, XGBoost and ELM attaining the accuracies between 84% and 98.3%. Studies like [18],[19]. highlighted the value of computational modeling for understanding smoking behavior and outcomes. Although these approaches have strong predictive performance, they largely lack interpretability, resulting in a research gap that justifies the use of XAI-based techniques like we do. Table 2.1 shows the literature review:

Table 2.1: Summary of Literature Reviewed.

Author(s)	**Year**	**Title**	**Methodology**	**Key Findings**
Shazzad et al.[9]	2013	Causes of Drug Addiction in Bangladesh	Survey	Identified socio-cultural and family-related causes; no AI used.
Khanam et al.[5]	2024	Drug Use Prediction with XAI	ML + LIME	Explained influence of stress and impulsiveness on addiction risk.
Arif et al.[7]	2021	Addiction Risk Prediction Using ML	ML models (97.9% accuracy)	High accuracy; lacks interpretability.
Bennett et al.[12]	2013	Lifestyle Factors in Student Drug Use	Logistic Regression	Linked nightlife and independence with drug use.
Tito et al[4]	2024	Early Detection of Drug Addiction Risk	ML models (95% accuracy)	Effective early detection; no explainability.
Zhang et al. [13]	2019	Drug and Alcohol Risk Prediction	Logistic Regression	Achieved 97.91% accuracy; strong predictors of substance risk identified.
Zhu et al. [10]	2017	Smoking Behavior Prediction	XGBoost	Achieved 84.11% accuracy; identified behavior-related smoking factors.
Zhant et al. [8]	2018	Tobacco Disease Detection	ELM	Achieved 98.3% accuracy in tobacco-related disease detection.
Mutamed et al. [14]	2025	Investigating youth drug addiction causes	Survey	Identified Addiction arises from multifactor influences.
Lee et al. [6]	2019	Smoking Status Prediction	Logistic Regression	Achieved 83.44% accuracy.

Kinreich et al. [2]	2019	Alcohol Use Disorder Treatment Seeking	Logistic Regression	Identified factors influencing treatment seeking (NM1).
Kumari et al. [1]	2018	Alcohol Use Disorder Risk Prediction	NM1	Highlighted key risk indicators (NM1).
Abdul Ahad et al. [13]	2022	Youth drug addiction causes in Sylhet	Survey	Abdul Ahad et al. [21]

2.2.1 Similar Applications :

Digital platforms, mobile applications, and research frameworks have developed utilizing data-driven or AI methods for the purpose of addiction prediction, monitoring as well as intervention. Such apps have methodological implications for this study.

- Machine Learning-Based Addiction Prediction SystemsEffective prediction of alcohol and drug abuse has been reported by studying models such as the Random Forest, SVM and ANN. While such systems showcase the might of AI in public health analytics, they are not interpretable, highlighting the importance of Explainable AI methods—like SHAP and LIME—that are used in this project to ensure that the decision-making is transparent.
- Quit Genius – Digital Addiction Rehabilitation App Quit Genius Quit Genius uses CBT and AI analysis to help people stop smoking, as well as promoting sobriety from alcohol and opioids. Through monitoring of behaviors and psychological statues, it resembles the multi-dimensional approach in our system which detect crucial factors for addiction.
- DRUGNET The drug abuse monitoring systemWeb-based early warning drug alert system, DRUGNET employs analytic methods to monitor trends and evidence potential outbreaks of drug use in communities. Its data-centric centralized monitoring system is consistent with our project's predictive modeling architecture.
- Online Tool Based on the Addiction Severity Index (ASI)ASI measures the severity of addiction in medical, psychological, social and environmental dimensions. Its structured and extensive, factor-based evaluation approach correlates well with our feature engineering method to detect socio-economic and psychological factors for addiction.

2.3 Gap Analysis

Most current research on substance use disorders is based on survey and statistical methods, which have little predictive power [9]. While machine learning models have demonstrated strong predictive power for addiction risk identification [7],[4]. they generally act as black-boxes, failing to disclose any information about how and why predictors affect predictions. Only a few studies utilize the Explainable AI, like but they have limited feature coverage and weak applicability in real scenarios [5].

The research on prediction of smoking and alcohol behavior is also promising, however these approaches do not have interpretability and none of them focus on the socio-cultural context in Bangladesh [15],[16].

In general, machine learning is lagging behind in XAI when neural network is used for interpreting the drug addiction of Bangladesh. In view of this gap, this study fills the gap by building an interpretable AI model with SHAP and LIME interpretation for practical, policy-related insight developments. The review of the literature is provided in Table 2 as :

Table 2.3: Gap Analysis of Existing Studies and the Proposed Research.

Features	Shazzad et al.[9]	Khanam et al.[5]	Arif et al.[7]	Tito et al.[4]	Zhu et al.[10]	Proposed system
Uses real-world data (Bangladesh)	Yes	No	Yes	Yes	No	Yes
Machine Learning applied	No	Yes	Yes	Yes	Yes	Yes
Explainable AI (XAI) used	No	Yes(LIME only)	No	No	Rare	Yes (SHAP + LIME)
Identifies root causes of addiction	Partially	Yes	Partially	No	Limited	Yes
High prediction accuracy	No	Yes	Yes	Yes	Yes	Yes
Transparent explanations	No	Limited	No	No	No	Yes
Focus on Bangladesh context	Yes	No	Yes	Yes	No	Yes
Multi-factor dataset (social + psychological + economic)	Limited	Limited	Moderate	Limited	Varies	Yes
Suitable for policy and rehabilitation decisions	No	Partially	No	No	No	Yes

2.4 Summary

- The general context of the issue was presented in this chapter, by describing what drug abuse entails and the sociopsychological and economic costs.
- The literature review presented summaries of key studies in addiction, smoking, and alcohol-related behaviors including these studies' methodologies and weaknesses.
- Analogous apps showed how machine learning has been used for similar behavioral prediction tasks in other populations.

Chapter 3

Research Methodology

This chapter presents the research methodology, system design, requirements, data flow, UI design, and project plan, along with task allocation and a brief summary.

3.1 Methodology

In this study, a cross-sectional survey approach is employed to examine the socio-economic, psycho-social and environmental factors of drug abuse. Information was obtained by means of a structured self-administered questionnaire on demographic information, psychologic factors, family and social factors, availability of substances and behavioral variables.

The study sample included 15–60 year old people from various socio-economic strata in urban and rural areas of Bangladesh drug users and non-drug users. The sample was purposeful for representation by demographic groups. It was a voluntary process and informed consent was taken from each respondent. Those with partial response or participants who withdrew during the data measurement period were not included in the analysis.

3.1.1Proposed Methodology

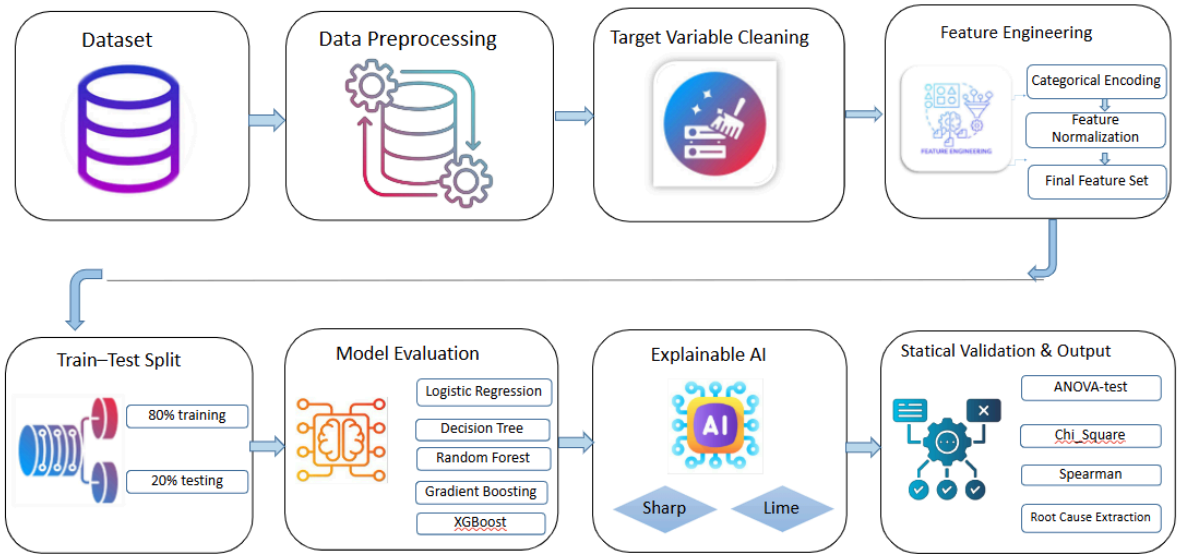


Figure 3.1.1: Research Design Framework

3.1.2 Overview

The research methodology is a structured, data-guided approach to make sense of the

root causes of drug addiction through statistical analysis, machine learning models and Explainable Artificial Intelligence (XAI). The general framework of the workflow aims for result accuracy and robustness, as well as explainability.

The procedure includes the following steps:

1. Dataset and Study Scope

- The raw dataset consists of 531 survey samples with 19 attributes including demographic, socioeconomic, psychological, environmental and sociocultural features.
- Only participants who responded Yes to Drug Use were retained for analysis.
- Of those, 362 were confirmed drug uses following the first filtering.

2. Data Preparation

- Demographic, behavioral, psychological environmental and socio-cultural attributes which are present in the dataset.
- Instead of dropping those records with missing values, they were replaced using the median for numerical features and mode for categorical features.
- The dependent variable (motivation to drug) was cleaned and labeled.
- Predictors were transformed into 49 binary features by one-hot-encoding.

3. Train–Test Split and Class Imbalance Handling

- 80:20 stratified train–test split was used to maintain class balance.
- Moderate class imbalance among addiction-reason categories was addressed by means of class weighting methods.

4. Statistical Analysis

- Categorical variables were compared using the chi-square test.
- Group comparisons were made using independent t-tests.
- Pearson and Spearman correlation tests for numerical variables were performed.
- Also, at ($p < 0.05$), peer pressure, psychological stress, family problems, love failure and economic pressure emerged as significant contributing factors outside of curiosity.

5. Feature Importance Analysis

- Correlation was used to study the linear relationships between variables.
- Mutual Information was used to detect non-linearities.
- Multicollinearity and feature stability were evaluated via the analysis of Variance Inflation Factor (VIF) .

6. Model Development

The trained machine learning models were the following:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

All models were trained in the same condition for fair comparison.

7. Model Evaluation

- The model was assessed by Accuracy, Precision, Recall, F1-score and ROC–AUC.
- Validation and stability testing were performed stratified.
- XGBoost came out as the most consistent and best model for this task.

8. Explainable AI (XAI) Framework

- The SHAP was utilized for global and class wise contribution analysis.
- We utilized LIME (Ribeiro et al., 2016) for example-level interpretability.
- The importance of each feature was confirmed using permutation importance and deep analysis.

9. Root Cause Extraction

The final root causes were determined by combining evidences from:

- SHAP rankings
- Mutual Information scores

- Permutation importance
- Statistical significance tests

Multilevel validation ensured robust identification of risk factors for drug addiction that are evidence-based.

3.1.3 Functional and Nonfunctional Requirements

Functional Requirements

1. **Data input from databases:** The system needs to be able to take data from the datasets of rehabilitation centers and online surveys. It would be nice to support well-known formats like CSV and XLSX.
2. **Data Preparation:** The dataset needs to be prepared; missing attributes handled, encoded for the categorical attribute, normalized for the numeric attribute and outlier smuggled off train set.
3. **Exploratory Data Analysis (EDA):** The interface shall construct descriptive statistics, charts, histograms: At the entry point, it has to act out a few simple EDA tasks such as clutter breaking, missing value checking and correlation testing.
4. **Statistical Analysis:** The system needs to perform statistical tests, such as t-test, chi-square and Pearson/Spearman correlation analysis, in order to detect variables that have possible significant association with drug abuse.
5. **Training Models:** I should train the model for various machine learning models like Logistic Regression, Random Forest and XGBoost. The seperation of data into training and testing sets should be supported.
6. **Modeling Evaluation**: The system should be able to calculate evaluation including accuracy, precision, recall, F1-Ievel score, ROC–AUC and confusion matrix. Colouring of results for comparison would be desirable.
7. **Explainable AI (XAI) Output:** The generated output must be explained using SHAP and LIME. It should present global explanations (”feature importance”) and local explanations (individual predictions).
8. **Root Cause Analysis:** The system should find and report those most determinant factors contributing to the likelihood of addiction among different category of people. And the more important, It must describe the main socio-economic, psychological and behavioral determinants.
9. **Subgroup Analysis:** The system needs to analyze patterns of the addiction in age, gender, location and employment difference, and compare them.
10. **Visualization of Results:** The system should show charts, diagrams and heat maps or SHAP graphics in a comprehensible way. It is only image products, which are focused on the interpretation requirements of policymakers and researchers.

Non-Functional Requirements

1. **Performance:** It should be able to handle data sets fast, and perform model training & explanation within acceptable computation time.
2. **Reliability:** The system needs to return the same answer on multiple runs. Mistakes such as bad inputs or missing files have to result in a safe, readable error message rather than a system hang.
3. **Usability:** Ease of use Interface and workflow should be easy, well-organized and beginner friendly etc. SHAP and LIME plots should be easy to understand also for non-expert humans.

4. **Modularity – Maintainability:** The code base of the system should be modular, with distinct modules for preprocessing, modeling and explainability. This will make possible the future changes, further models, or larger data sets.
5. **Confidentiality:** All sensitive data must be safeguarded by the system. Any personal identifiers should not be retained, displayed or transmitted thus protecting the anonymity of the survey participants.
6. **Portability:** It should be executable on common platforms like Windows, Linux, cloud. It should work seamlessly in Jupyter, Colab, or other Python IDE.
7. **Scalability:** Systemia should be capable of scaling for larger datasets, more features, new algorithms or XAI methods to be studied in the future.

3.2 Detailed Methodology and Design

In this section, we describe the complete methodological framework used to build our drug addiction prediction system. We also discuss several alternative approaches that were explored during development. Ultimately, the chosen design was guided by three main considerations: how consistently the model performed across different settings, how clearly the explainable AI techniques (such as SHAP and LIME) could justify each prediction, and how well the models classified risk in real data. By balancing accuracy with transparency, the proposed system aims to provide meaningful support for decision-making in addiction prevention and public health interventions.

3.2.1 Alternative Methods Considered

a. Alternative Machine Learning Models

These models were chosen due to their well reported performance for classification. Yet, all had limitations for the purposes of the current study. SVM is expensive to the computation when scaled up in terms of both size and types, KNN does not work well when applied on high-dimensional numerical features mixed with categorical values, LightGBM as a fast model lacks interpretability, ANN operates like a “black-box” thus failing to reveal root cause.

b. Other Explainable AI (XAI) Approaches

Several other XAI techniques including LIME, Permutation Feature Importance (PFI), Partial Dependence Plots (PDP), and surrogate decision trees were also considered. But these methods did not satisfy our requirement of interpretability in this study. lLIME gives unfathomable local explanations, PFI does not account for interactions, PDP tends to show oversimplified non-linear behaviour and surrogate models do not capture all the features of the orignal model diminishing explanation quality.

c. Alternative Data Collection Strategies

Various data collection methods were considered such as secondary data sources, online-only surveys, and brief cases studies. These methods were deemed inapplicable as they do not take into account socio-economic and psychological factors which are significant for Bangladesh. Secondary data sets do not have local contextualization, online surveys risk selection bias and small case studies are not generalizable. Therefore, an original dataset mixing rehabilitation center data and structured surveys was chosen as the most healthy-related source for this purpose.

3.2.2 Justification for the Selected Methodology

The last method, which is a combination of the RF logistic and XGBoost models with explainability based on SHAP and LIME, was chosen as giving the best compromise between accuracy, interpretability, and implementation.

Random Forest and XGBoost were chosen as the few models that manage mixed data type well, capture non-linearity of relationships and give stable and accurate predictions for addiction risk. To provide interpretations, we apply both SHAP and LIME. SHAP provides interpretable global explanation and understandable feature interaction insight, while LIME generates understandable and easily accessible case-level interpretations. Applying both approaches leads to a more comprehensive and illustrative view on how particular sources contribute toward the risk of addiction.

We selected our primary data from the rehabilitation centers and the survey since it represents the actual social, economic and psychology aspects of Bangladesh. The benefit of the YGCS is that information such as peer pressure, family conflict and emotional stress which may be specific to our context are not available in foreign or secondary data sets. This ensures that the findings are more robust and context-relevant for policy makers or rehabilitation programs.

3.3 Project Plan

Developing, analyzing and evaluating WHILE projects The project was implemented according to the phased plan of system design. The work is structured in stages: after the study of requirements and data set preparation, system design, model development, XAI integration have been developed, ending with evaluation and reporting. Each phase has specific goals, outputs and timings to ensure continuity and momentum through the project.

Phase 1: Requirement Analysis

- Identify system requirements.
- Define functional and non-functional requirements.
- Know the problem domain and your application as a whole Since component documentation can quickly become difficult to manage pamphlet bestseller length, managing visual patterns across an entire app is near impossible.

Phase 2: Data Collection & Pre-processing

- Dataset collection (survey and data from rehabilitation centers).
- Clean and preprocess data.
- Deal with missing data, outliners and imbalance

Phase 3: Exploratory Data Analysis

- Do some basic EDA (demographics, target variable). Performing statistical EDA (ANOVA-test, Z-test, Chi-square, correlations).
- Generate initial insights

Phase 4: Model Development

- Split dataset.
- Fit few ML models (LR, RF, XGBoost).
- Hyperparameter tuning and cross-validation

Phase 5: Explainable AI Integration

- Use SHAP for global and local interpretations.
- Apply LIME for case-wise interpretation.
- Compare and validate explanation consistency

Phase 6 : UI designing and System Layout

- Develop system architecture
- Create UI wireframes
- Ensure usability and workflow clarity

Phase 7: Evaluation & Validation

- Assessment of the model performance (AUC, F1 score, and CM)
- The significance of features in results is confirmed with the help of statistical inferences.
- Conduct subgroup and root cause analysis

12Phase 8: Documentation & Final Report

- Write research methodology, results and discussion
- Write conclusion and future work
- Final proofreading and submission

3.3 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 12 to week 48.

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Data collection phase																			
Preprocess all the data																			
Feature Analysis																			
Model training																			
XAI																			
Statical Validation																			

3.4 Summary

This chapter opened with a discussion of the research methodology that was as a whole used to achieve objective and analytical purpose statements. We aimed to develop this approach in order to find and explain risk factors for drug addiction based on a combination of data statistics, machine learning models and Explainable Artificial Intelligence (XAI). The strategy taken ensured a trade-off between predictive accuracy and model interpretability, which is important for capturing human-interpretable information in complex data.

The chapter also provided the methodological and system development aspects, such as data preprocessing, feature construction, model selection and evaluation methods. In addition, more specific attention was drawn to explainable AI (XAI) methods such as SHAP and LIME for the purpose of increasing transparency/explainability of model predictions. Other analytical methods have been initially planned to be used, but the selected ones were determined to be appropriate in this study based on appropriateness, rigor, interpretability and potential of addressing the research questions. This was to ensure that the analytical approach would remain methodologically rigorous and appropriate for the identification of risk factors for addiction.Finally, we sketched the planned project structure by detailing the subsequent stages of data preparation, modelling and explainability, to overall documentation. The systematic structure allowed positive work coordination as well as a similar mode of the research at different stages. In general, the model discussed in this chapter provides a coherent, well-reasoned and transparent catalyst for seeking to achieve .

Chapter 4

Implementation and Results

This chapter presents the implementation of the system and the analysis of results obtained from model training and evaluation, including performance metrics, confusion matrices, and comparative model outcomes.

4 Environment Setup

Experiments are carried out on an ASUS X415 laptop with Intel Core i5 and 8GB RAM under Windows 10 (64-bit). Tab. 2 summarizes the performance of the system to accomplish data preprocessing, statistical analysis and machine-learning model training as well as XAI tasks. All code was developed in Python 3. x with Jupyter Notebook and Google Colab to enable users fully local or cloud execution.

The main libraries utilized in the present study were mainly composed of NumPy, Pandas, Matplotlib and Seaborn for the manipulation and visualization of data, Scikit-learn and XGBoost for building machine learning models, SHAP and LIME to analyse explainability. This setting allowed for the successful performance of all parts of analyses necessary to classify the underlying factors of drug dependence. Table 4.1 shows the used Tools & Environment:

Table 4.1: Experimental Setup .

Category	Tools/Details Used
Operating System	Windows 10 (64-bit)
Development Platform	Google Colab
Programming Language	Python 3.x
ML Frameworks	Scikit-learn, XGBoost
XAI Libraries	SHAP, LIME
Supporting Libraries	NumPy, Pandas, Matplotlib, Seaborn
Statistical Tools	SciPy
Runtime Environment	CPU (Intel Core i5, 8GB RAM)
Data Storage	Local Drive + Google Drive

4.1 Demographic and Socio-Economic Characteristics of the Study Participants

4.1.1 Age Distribution of the Respondents

The age distribution of participants in the study is shown in Figure 1. The distribution indicates that there is a peak in the population at relatively young ages.com)Age was calculated as 2019 minus the birth-year we collected from source; for education, which includes all diplomas/degrees a participant may have obtained, four categories were recoded. The bold vertical line represents the mean age of the respondents: 25.7 years. This also implies that drug dependence is more frequent among younger people in the population of the study and indicates early adulthood as a moment of higher vulnerability. Fewer participants are from older age groups and are thus comparably underrepresented in the aged. This 4.1.1: Age Distribution of Respondents:

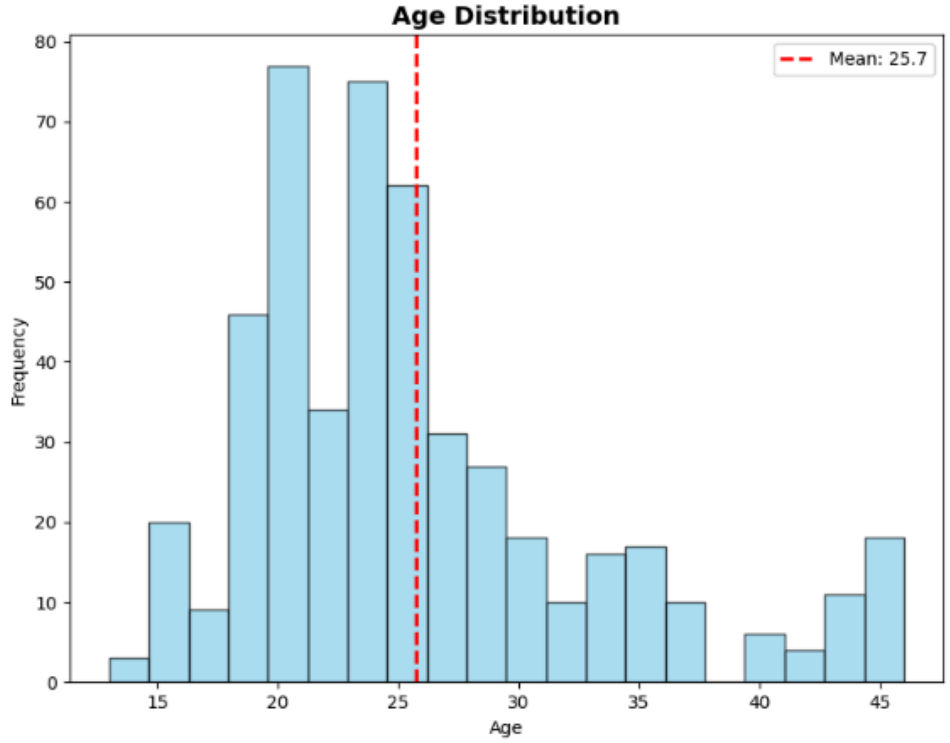


Figure 4.1.1: Age Distribution of the Respondents

4.1.2 Gender Distribution of the Respondents

The sex disaggregated population of the study sample is reflected in Figure 2. The majority of the dataset is male-driven, where males are responsible for most of the cases. There are relatively few women represented here, and to those from other gender-groups the number of respondents is negligible. This unbalance indicates that a large proportion of the drug addiction cases presented in our dataset are associated with male subjects, but that could be due to social exposure (which would affect reporting) or risk behaviors. Figure 4.1.2: Gender Classification of the Respondents:

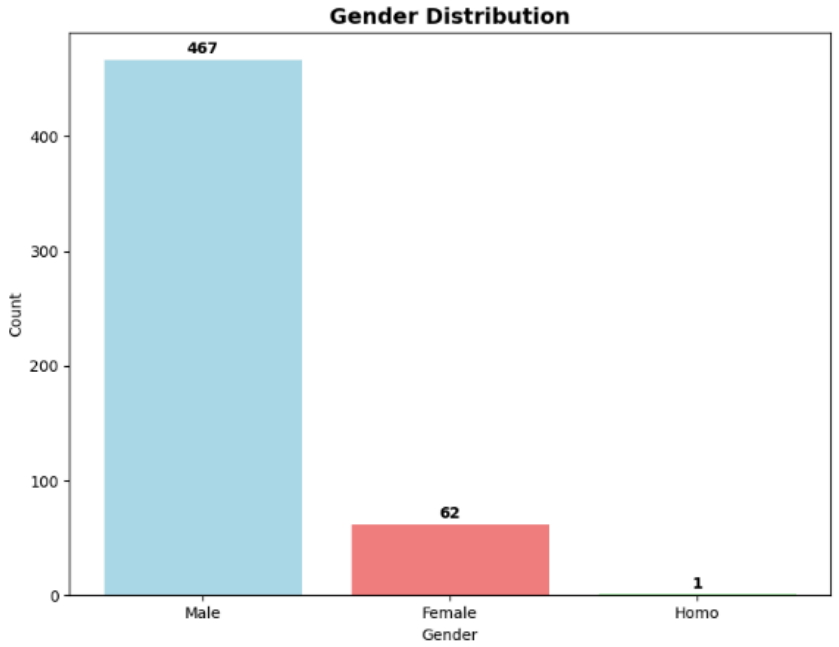


Figure 4.1.2: Gender Distribution of the Respondents.

4.1.3 Location Distribution of the Respondents

the residence of the respondents. The data show a fairly even split between respondents from urban and rural areas, and slightly more of them are from urban areas. Evidencing that love for drugs is not just in one place but belongs to both cities and villages, whose hands it has continued patching its patch-patches on. Location Distribution of the respondents: Figure 4.1.3



Figure 4.1.3: Location Distribution of the Respondents

4.1.4 Distribution of Employment Types

Figure The most represented job categories of the respondents. The pattern shows that students are the most in number, self employed and unemployed next. A smaller percentage are entrepreneurs and professionals. The student-dominated prevalence also emphasizes the susceptibility of younger and academically engaged at university population towards drug dependence, whereas there is unemployed individuals on the other hand novels possible relationships between unemployment-levels and substance abuse. This Figure 4.1.4: Composition of Employment Category:

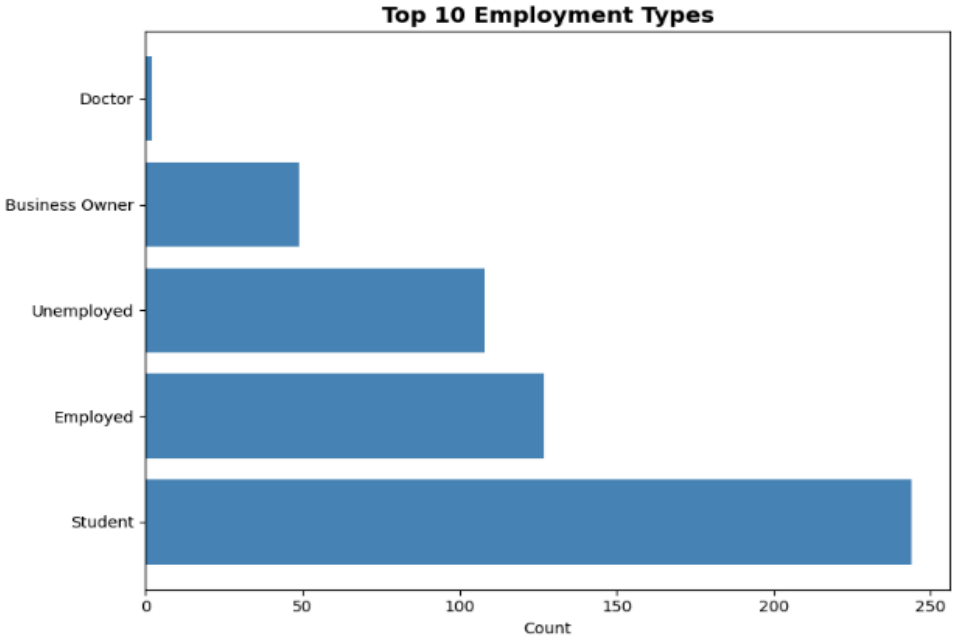


Figure 4.1.4: Distribution of Employment Types.

4.1.5 Drug Use Distribution

Fig. 1 describes the distribution of the respondents' drug use status among those who were included in the data set depicted. As a result, drug use was recorded in the case of 68.3% (the majority) sample-materials and in 31.7%-no consumption participants. This imbalance shows that the dataset is mostly made of drug-experienced subjects, and can be exploited to explore the causes and effects associated with addiction.

The relatively high rate of drug users also supports further analysis focusing on risk factors and root causes of drug addiction based on machine learning algorithms and Explainable AI methods. Meanwhile, the existence of a non-user group allows contextual contrast and facilitates preliminary evidence as well as comparative studies. This 4.1.5 Figure Use of Drug:

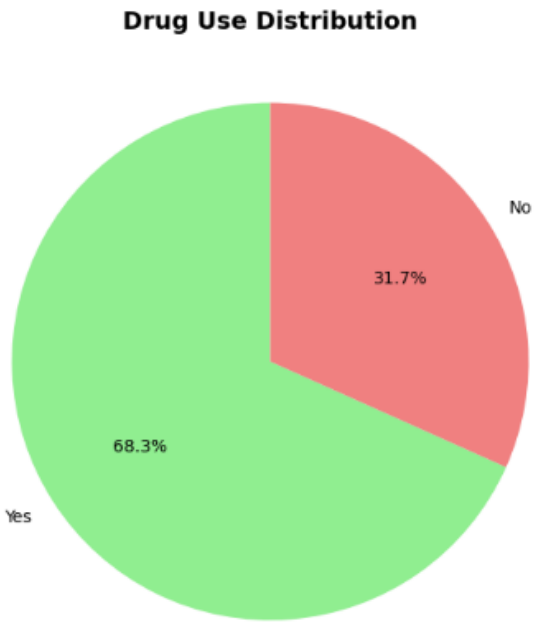


Figure 4.1.5: Drug Use Distribution.

4.2 Testing and Evaluation/Performance/ Comparative Analysis

4.2.1 Spearman Correlation :

The Pearson correlation heatmap demonstrates linear relationship of numeric variables across demographic, psychological, emotional (emotional–regulations), financial and socio-culture attributes. An Age and Start Age have a strong association (r = 0.68), so that those who are older now started their drug use later in life. Psychological stress in general has moderate positive correlations with Life Failure (r = 0.57), Love Failure (r = 0.46), and Financial Pressure (r = 0.44); i.e., emotional and economic problems are strongly involved in enhancing psychological stress. This Figure 4.2.1: Correlation Heatmap of Numerical and Categorical Encoded Features:

Financial Pressure is also moderately associated with Love Failure and Cultural Pressure, which indicates it intertwines socio-economic and emotional problems. On the other hand, Family Relation and Satisfaction show low correlations with nearly all variables suggesting that their relationship to addictive behavior might have been indirect or not linear.

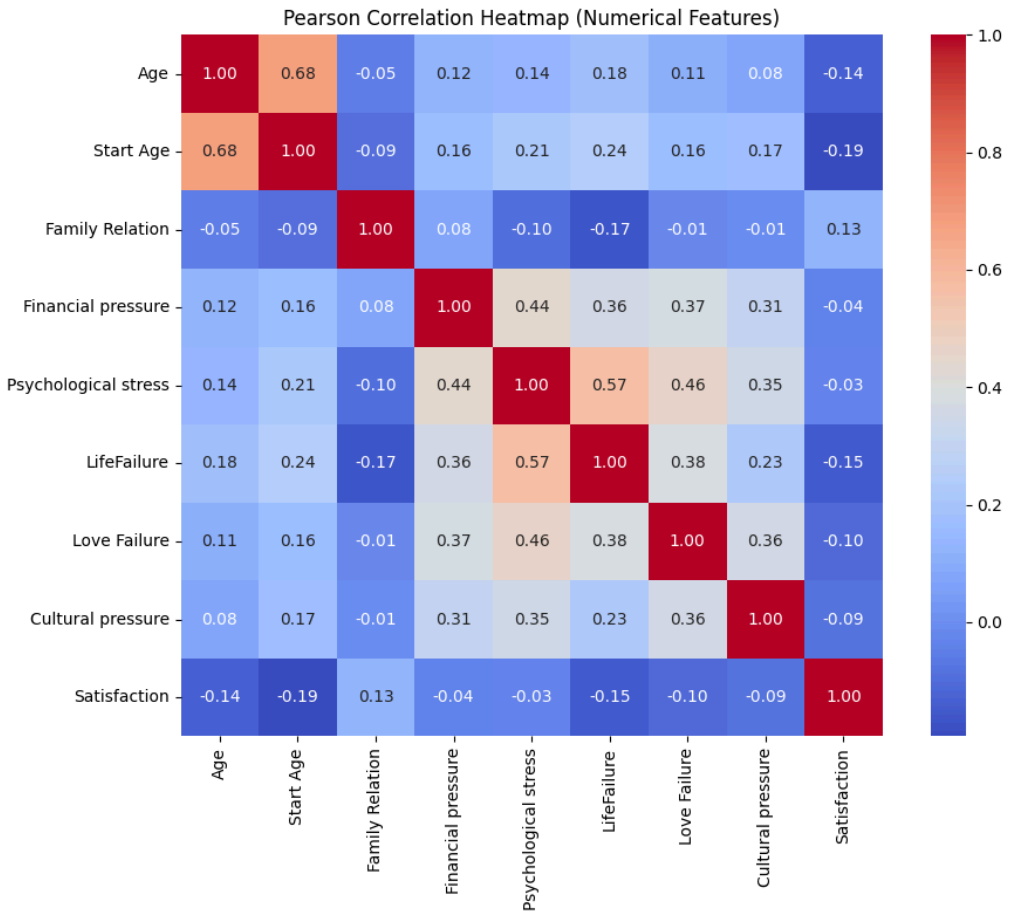


Figure 4.2.1: Correlation Heatmap.

In general, the heatmap shows a lack of strong linear correlations among majority of features, justifying the use of non-linear machine learning and explainable AI methods to model these intricate process leading to drug addiction.

4.2.2 Mutual Information (MI) Scores :

The MI scores measure both linear as well as non-linear relationships between the variables and the goal. The MI is higher for Age and Start Age, thus indicating that factors associated with age may be the most informative features in predicting reasons for drug addiction. : This Figure 4.2.2 Mutual Information Scores of Drug Use Prediction:

Psychosocial factors, including life.s failure; family relation; cultural pressure, psychological stress with MI scores are also quite high that indicate significant themes for safety. In contrast, numerous fine-grained categories occur with very low MI values also hinting at an indirect contribution.

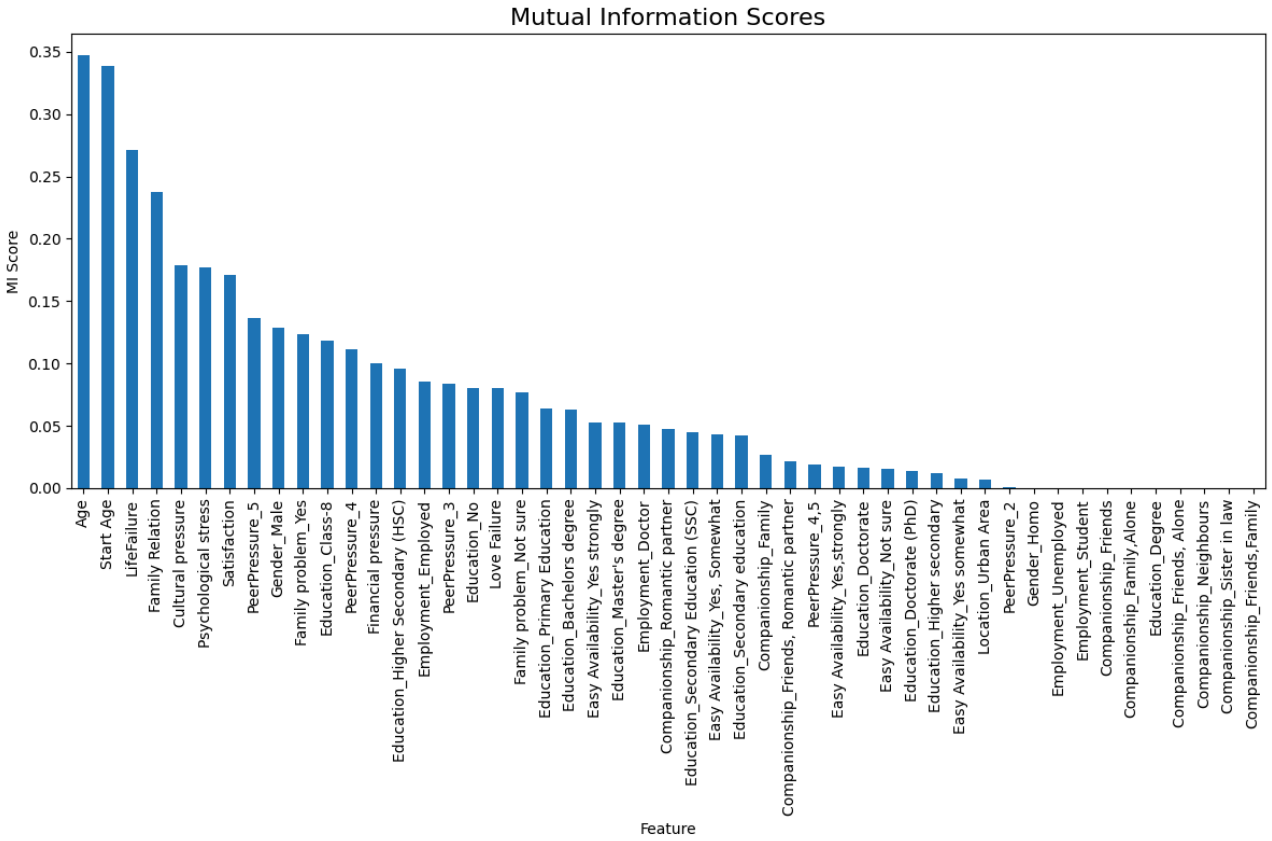


Figure 42.2 : Mutual Information Scores .

Taken together, the results provide evidence of non-linear complexity in drug addiction and thus: (i) corroborate the rationale for incorporating non-linear machine learning and explainable AI methodologies.

4.2.3 Model Performance :

The prediction results of each model showed in the Table, include accuracy, weighted precision, recall and F1-score.. Figure 4.2.4 is shown all Machine Learning Models Performance:

Table 4.2.4 Performance Summary of All Machine Learning Models.

Model	Accuracy	Precision	Recall	F1-score
XGBoost	0.7945	0.7908	0.7945	0.7817
Random Forest	0.7671	0.7540	0.7671	0.7427
Decision Tree	0.7534	0.7331	0.7534	0.7307
Gradient Boosting	0.7397	0.7441	0.7397	0.7262
Logistic Regression	0.5342	0.5528	0.5342	0.5185

Among all models, XGBoost obtains at the best performance in terms of all evaluation measures and has an accuracy and weighted F1-score of 0.79 and 0.78, respectively which suggests that it demonstrates strong classification capabilities while balancing out precision in the multi-class scenario. Random Forest and Decision Tree also show excellent performance, the results are a little below than those reached by XGBoost as they can capture non-linear properties. Gradient Boosting has a fair performance, and Logistic Regression has very bad results, and thus it underlines the inability of linear models to deal with this complex problem. On the whole, they argue for ensemble non-linear models being more appropriate in predicting reasons for drug addiction.

4.2.5 Performance Comparison :

In this Figure 5, the performance comparison of five machine learning model with their accuracy, weighted precision, weighted recall and weight F1-score is visualized. XGBoost achieves the best performance than all other models on all metrics in the multi-class classification and its predictive power is relatively even. This Figure 4.2.5 illustrates Performance Comparison for reasons analysis :

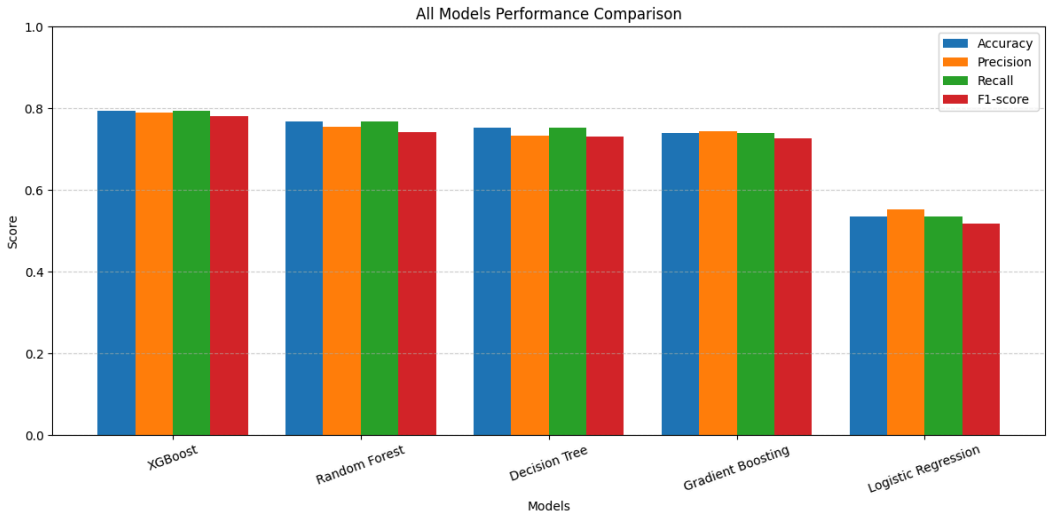


Figure 4.2.4 : Performance Comparison bar diagram.

Random Forest and Decision Tree also perform well (albeit worse than XGBoost), which may be attributed to their capacity to model non-linear data relationships. Gradient Boosting achieves similar but slightly lower results and Logistic Regression yields remarkably less successful overall performance, indicating that the modeling of patterns related to addiction cannot be effectively captured by linear models. All in all, the graphic provides clear evidence that ensemble, non-linear models outperform classical linear ones, thus explaining the choice of XGBoost as the final model for additional explainable AI domain analysis.

4.2.5 Confusion Matrix Analysis (XGBoost) :

The results of class-wise prediction by the XGBoost model to classify reasons for drug addiction are given in the confusion matrix. High diagonal true positive values mean a good performance in classification of dominant9 categories such as Peer Pressure, Curiosity, Failure in Love, Lack of Awareness and Loneliness or Depression. Figure 4.2: Confusion Matrix Analysis (XGBoost).

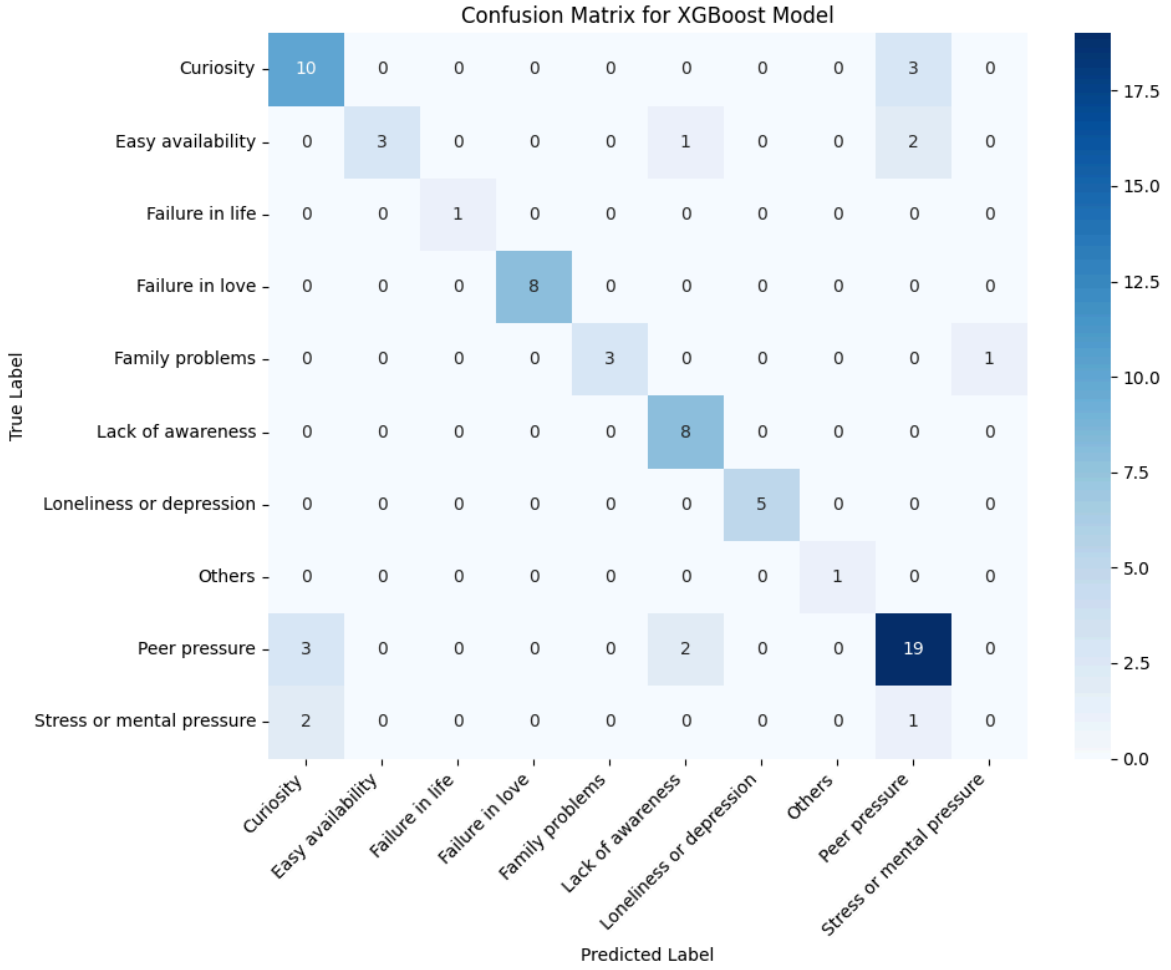


Figure 4.2.5: Confusion Matrix Analysis (XGBoost).

There are few misclassifications, mainly between socially and psychologically related classes like Peer Pressure, Curiosity, Stress or Mental Pressure indicating such reasons shares common characteristics. Easy Availability and Peer Pressure also exhibit slight confusion (A4) which however mirrors real life interactions among social influence and drug availability. Consequently, the confusion matrix indicates that multi-class performance is powerful and well balanced, which ensures the soundness of XGBoost model for explainable analysis.

4.2.6 ROC Curve Analysis :

This Figure presents the macro-average Receiver Operating Characteristic (ROC) curves of all tested machine learning models for drug addiction reasons multi-class classification. The macro-average ROC reflects the discrimination power of each model over all classes, since it equally weights all classes and is appropriate to discriminations with class imbalances. The broken diagonal line and the embedded diagram represent a random classifier (AUC = 0.50).. Figure 4.2.6 illustrates All models ROC Curve AnalysisÊ:

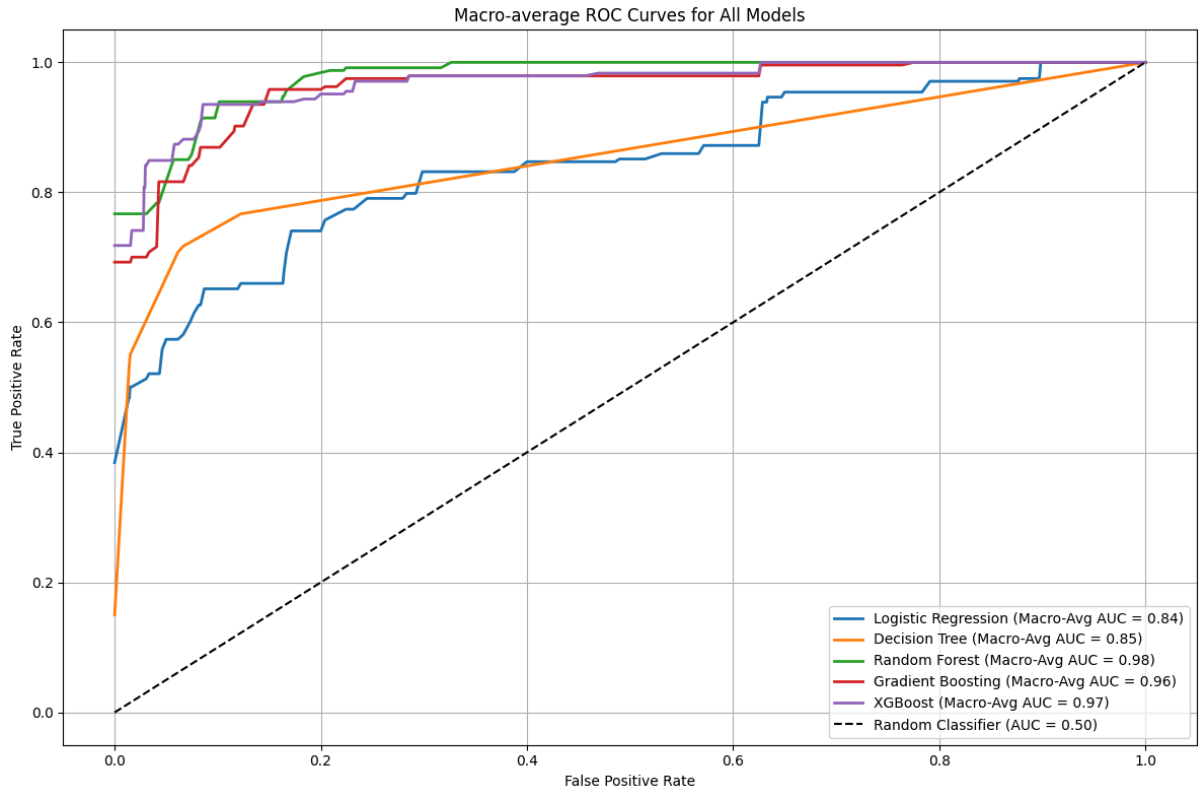


Figure 4.2.6 All models ROC Curve.

Of all the models considered, Random Forest has the best macro-average AUC (0.98), followed closely by XGBoost (0.97) and Gradient Boosting (0.96), ie high degree of perfect separation between classes. However, Decision Tree and Logistic Regression have lower AUC values (0.85 and 0.84) compared to SVMs, indicating less separation between addiction types for these methods. In general, ROC curves suggest that ensembles perform much better for discrimination than linear and single tree methods. These findings also confirm the choice of XGBoost as a trusted and powerful explainable model in this research.

4.2.7 Cross-Validation Performance Analysis :

This Table provides an overview of the cross-validation performance for all examined machine learning algorithms in terms of mean and standard deviation values over all folds. This validation provides a strong framework to estimate model stability and generalization because it avoids some effect from partition bias. Table 4.2.7 Summary of Performance on 10-Fold Stratified Cross-Validation:

The highest mean accuracy (0.8094) and F1-score (0.7940) of RF with low standard deviation suggests that it possesses strong predictive ability as well as being stable in most folds. Additionally, gradient boosting and XGBoost also appear to perform well with slightly lower mean scores but reasonable dispersion, although slightly worse than traditional models. These results draw the attention on their natural ability to find non-linear patterns in the data as a consequence of these methods based on aggregation of decision trees. The decision tree is the most high-variance model, indicating that it may be fitting too much to noise within the training set; and logistic regression has low performance on all of the measures, accounting for its inability to capture the complex non-linear associations between reasons of drug addiction.

Table 4.2.7: Cross-Validation Performance .

Model	Accuracy (Mean ± Std)	Precision (Mean ± Std)	Recall (Mean ± Std)	F1-score (Mean ± Std)
Random Forest	0.8094 ± 0.0165	0.8204 ± 0.0330	0.8094 ± 0.0165	0.7940 ± 0.0192
XGBoost	0.7791 ± 0.0317	0.7726 ± 0.0486	0.7791 ± 0.0317	0.7652 ± 0.0338
Gradient Boosting	0.7708 ± 0.0180	0.7699 ± 0.0348	0.7708 ± 0.0180	0.7596 ± 0.0189
Decision Tree	0.7597 ± 0.0416	0.7807 ± 0.0519	0.7597 ± 0.0416	0.7548 ± 0.0440
Logistic Regression	0.5193 ± 0.0352	0.5130 ± 0.0360	0.5193 ± 0.0352	0.4944 ± 0.0387

In conclusion, the cross-validation scores clearly demonstrated that models based on ensembles provide better generalization performance than those based on linear and single-tree algorithms which makes them appropriate for this multi-class classification problem.

4.2.8 Model Stability Test Across Multiple Random States :

The stability analysis of the XGBoost model tested on different random states is presented in Table X. We show that the variation in random initialization does not greatly affect the performance of XGBoost in terms of accuracy and macro F1-score. Random states 10 and 40 show the best performance, where the accuracy is 0.79, and the macro F1-scores are approximately equal to 0.79. Although we see a moderate level of performance fluctuations at random states 20 and 30, the deviation as a whole remains acceptable indicating stable generalization behavior. Table 4.2.8: Model Stability Test with Different Random States:

Table 4.2.8: Model Stability Test .

Random State	Accuracy	Macro F1-score
10	0.7945	0.7891
20	0.7397	0.7318
30	0.7123	0.7108
40	0.7945	0.7841
50	0.7671	0.7676

The macro F1-score has the same behaviour as accuracy, It indicates that our score is balanced for all classes and brings up again our hypothesis about the model does not favour majority classes taking random samples.

4.3 Results and Discussion

This chapter provides a detailed interpretation of experimental results from machine learning models along with explainable AI approaches and statistical validation. A comprehensive discussion on the numerical results of model performance, robustness and interpretability as well as an assessment for identifying root causes of drug addiction is presented. Comparisons with several classifiers have shown that ensemble learning models are significantly better, through the cross-validation and stability experiments we also verify the generalization ability. In addition, explainable AI approaches offer clear interpretations of the most important demographic, psychological and psychosocial factors during a pandemic. In this section, we combine the above results with some statistic information to guarantee validity and interpretability/ practical significance of the proposed scheme.

4.3.1 Dataset Characteristics and Class Distribution

The last 362 cases, who only consider drug users were retrained and used as the final data set. The summary of the entire dataset configuration is shown in Table 1.

The final dataset analyzed in this study comprises 362 samples after exclusion of those individuals who declared drug use. The dataset is initially consisting of 17 features and they are transformed into 49 encoded features after encoding the categorical data. The target variable is the cause of drug addiction, which has 10 different classes and shows that this problem is multi-class. The dataset is split into 80% train and 20% test ensuring that there is enough data for training the model while having an independent set to evaluate the performance.Figure 4.3.3 show Dataset Characteristics and Class Distribution :

Table 4.3.1: Dataset Description

Item	Description
Total samples (after filtering)	362
Total features (before encoding)	17
Encoded features	49
Target variable	Reason for drug addiction
Number of target classes	10
Train–test split	80% training, 20% testing

4.3.2 Distribution of Drug Addiction Reasons

Frequency distribution On the table, it can be seen that frequency distribution of a reason for drug addiction in the dataset. The peak reason is peer pressure, due to which the most number of children are introduced to the habit followed by curiosity. In addition, emotional and attentional aspects like suffering from love or unawareness stand for a remarkable percentage of the data. Reasons that are stress or pressure related, failed in live and others seem not to be that common. Figure 4.3.2 present Distributionthe of Causes of Drug addiction:

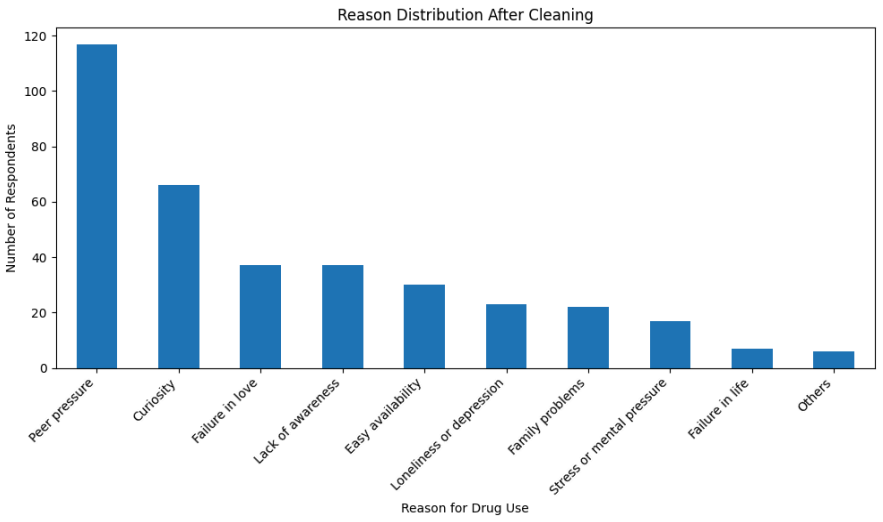


Figure 4.3.2:Distribution of Drug Addiction Reasons.

This imbalance reflects the social and exploratory aspects of drug addiction, and reveals an imbalanced class that is typical for real-world behavioral data that we take into account when training the model.

4.3.3 Selecting the Best Model for XAI :

Choosing a suitable model for explainable AI (XAI) is in general a trade-off between predictive accuracy, stability and interpretability. Some of the models were compared in this study through conventional performance measures. Linear methods turned out to be of little help; tree-based ensembles performed better.

Of all models, XGBoost always obtained high accuracy and balanced F 1-scores with more stable results through random states. Moreover, XGBoost can use for interpretable model gain-feature attributions methods like SHAP and LIME that provide good and trustable global as well local explanations. Thus, in view of all factors including performance, robustness and explainability, XGBoost was chosen as the final model for the XAI based analysis of this study.

4.3.4 SHAP Global Explanation :

SHAP Global Featur We provide the class-wise global feature importance computed using mean absolute SHAP values, where a small hamstringing the model to make predictions as independent of so acting as possible.SHAP summary plot for each disease.coreding corpus. We find Start Age to be the most important factor in all three cases, illustrating that an early start is predictive of which addiction pathways a user will pursue. Age, Love Failure and Life Failure also making contributions significant which support that the life stage and emotional stress play key roles in addictive behavior. It is found that in the majority of classes the following predictors are related to at least one class: Cultural Pressure, Financial Pressure and Family Relation highlighting the importance of social environment and family dynamic. Psychological Stress also adds to a number of types by fortifying the relationship between mental health problems and substance use. Domain-specific effects are also observed in categorical factors such as education, strength of peer pressure and ease to obtain cigarettes, which have class-specific effect patterns, meaning that different reasons for dependence occur through different combinations risk factors. Figure 4.3.4 show SHAP Global Feature Importance.

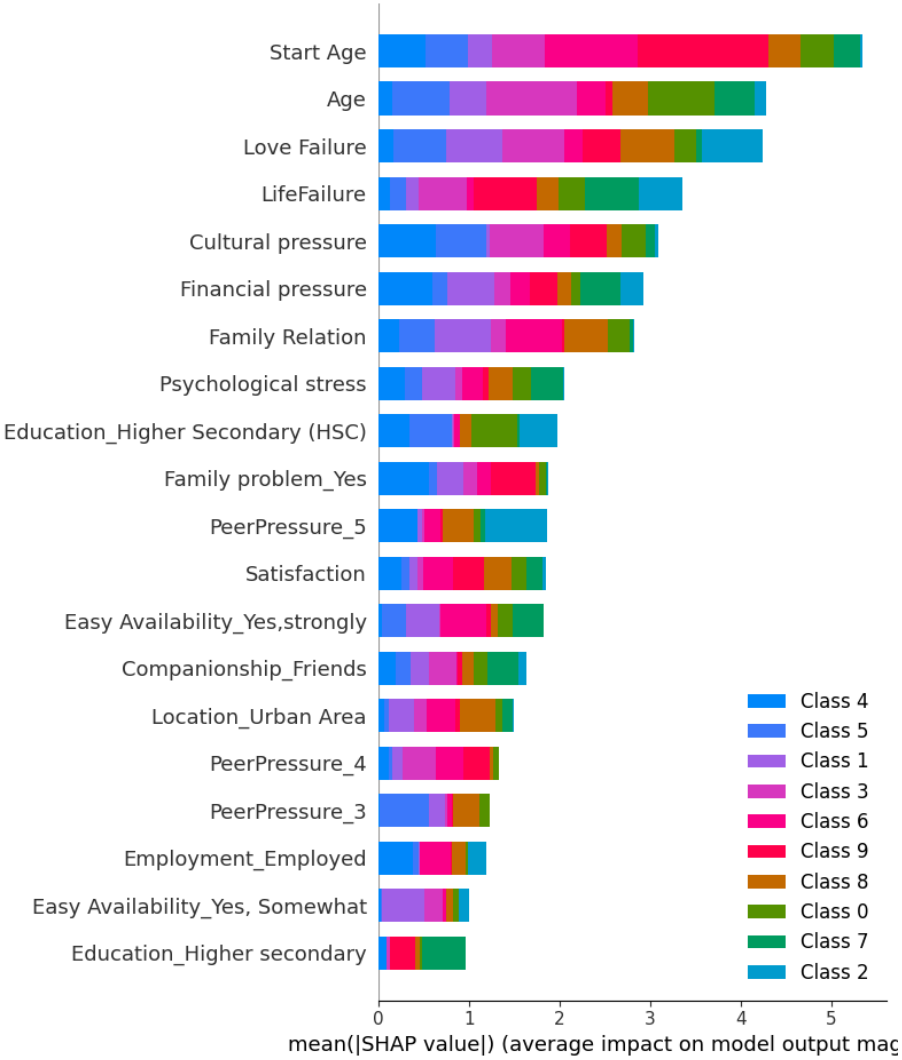


Figure 4.3.4: SHAP Global Feature Importance .

Overall, the SHAP analysis supports recent arguments that drug addiction is a multifactorial process with complex class-dependent feature correlations that can be effectively recognized and "explained" when models are produced using explainable AI techniques.

4.3.5 SHAP Dependence Plot Interpretation :

The SHAP dependence plots show which important features shape the prediction of curiosity-driven addiction class. Start Age exhibits a positive non-linear interaction, such that it positively moderates the relationship between type-I addiction and curiosity based addiction for higher levels of the start age and negtive ones when start age is lower. A similar pattern emerges for Age, suggesting that addiction to curiosity comes forward as people age. A different pattern is observed however for Love Failure, where high and low levels have mixed effects on the factor (negative generally), this might mean that curiosity motivated behaviors tends to be less related with extreme negative states despite emotional distress at higher intensity. On the other hand, Life Failure seems to have a sharper negative loadings in higher (front-to-back) values, what implies that people suffering from high life failures become less motivated by curiosity. This Figure 4.3.5 is dispaly SHAP Dependence.

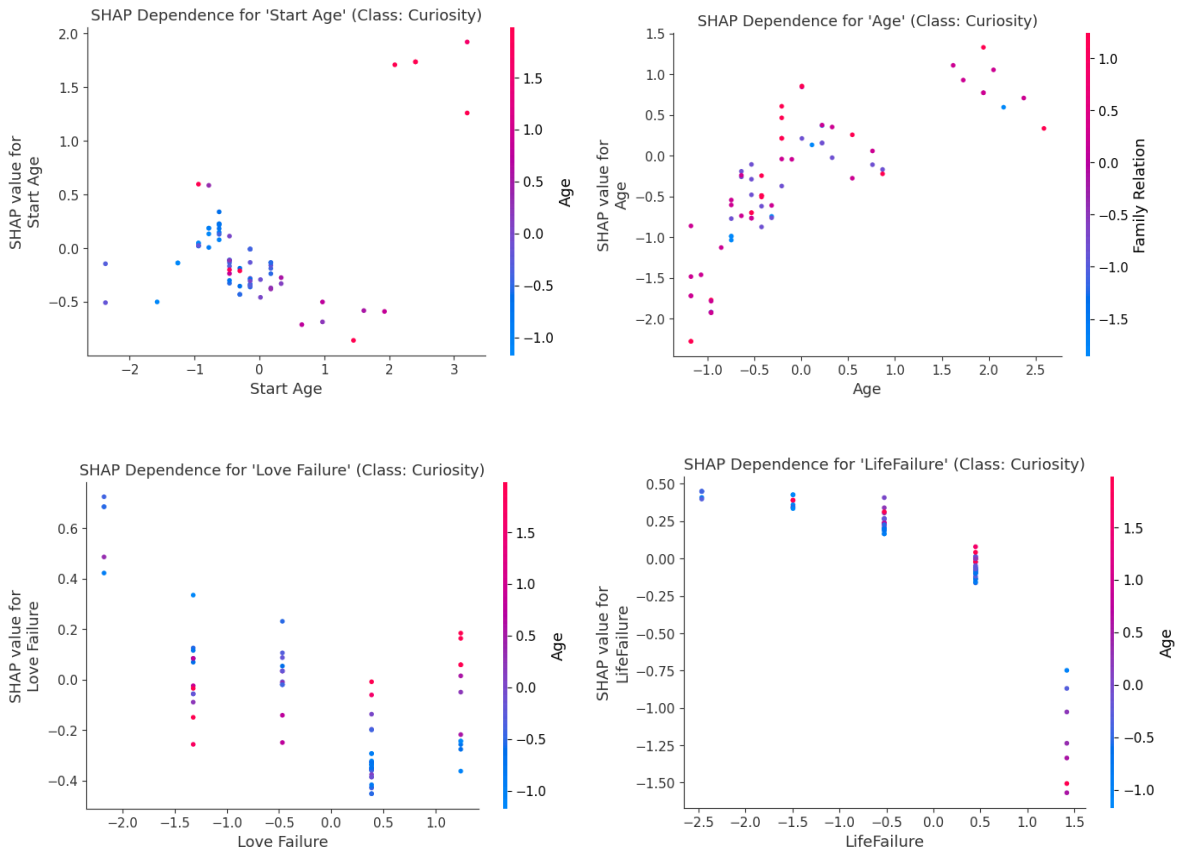


Figure 4.3.5: SHAP Dependence Plot .

Taken together these dependence plots show that curiosity-based addiction is primarily associated with explorative behaviors when age related rather than severe emotional or life disruptions, in line with the class-specific and non-linear patterns identified by the explainable AI analysis.

4.3.6 Class-wise SHAP Feature Importance Analysis :

For the proposed method, prediction explanations based on classwise feature importance analysis are shown in Fig.

The figure shows the top contributing features on the four major classes of reasons for drug addiction and how different feature exert impacts to them based on their mean SHAP values. In the FFM group of Curiosity, age factors including Age, Start Age and education are most important to determine curiosity-driven addiction -meaning that how addiction contributes to individuals who in mid-life stage where they have more knowledge exposure; as well as not distressed them very high (and consistent with Factor II above). Family Relation, Love Failure, Financial Pressure and availability are the significant factors more prevalent in Easy Availability class depict addiction of subjects from this category was more or less like a

triangle spread of extended Five years support pointed to family relation, loved failure may leads individual towards loneliness underneath caste, creating obdurate space under the family's cover with echoes created by them. Social Pressure, Love Failure and Life Failure serve as strong contributors in the Failure in Life class, mirroring addictions pathways based on peer pressure addiction and successive life falterings. For the love failure class, Age, Love Failure and Cultural Pressure AND Start Age are among the crucial contributing factors further stressing on the role of emotional and social-cultural setting. This Class wise SHAP Feature Importance Analysis is Shown in above Figure 4.3.6

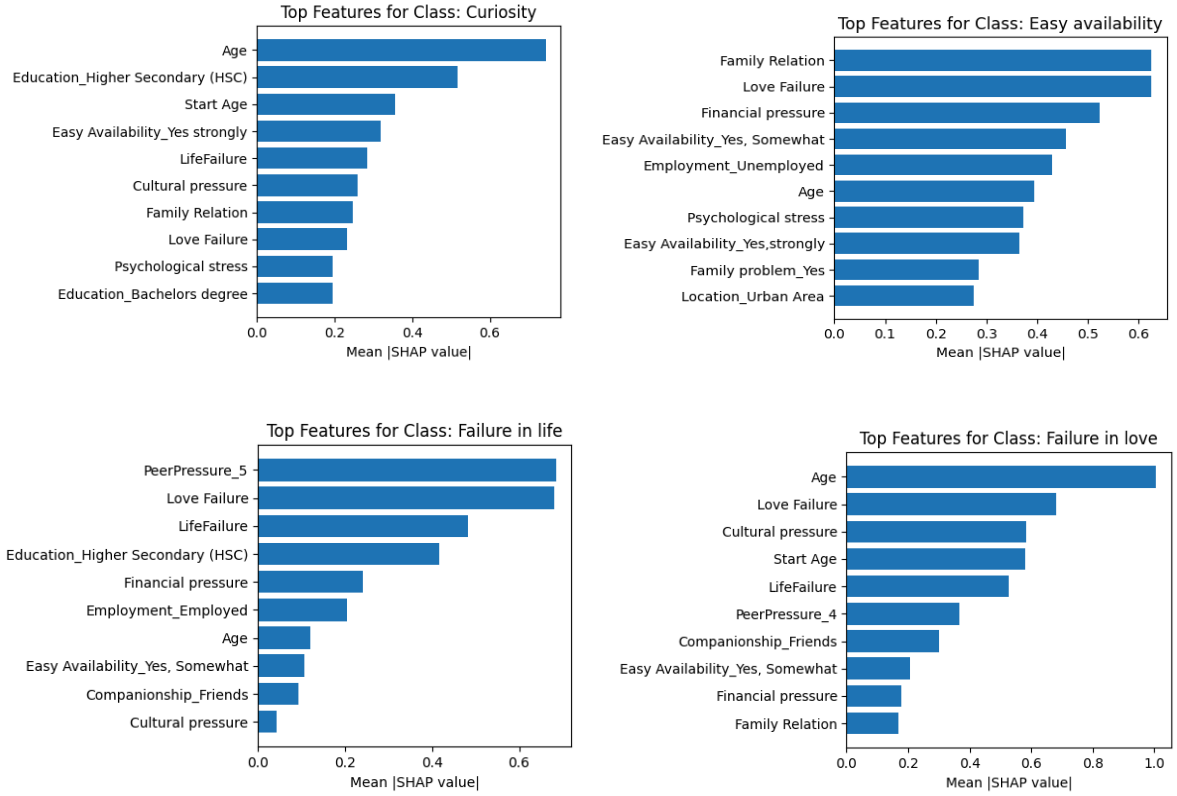


Figure 4.3.6: Class-wise SHAP Feature Importance Analysis .

In summary, this class-wise SHAP analysis is illustrative of the fact that drug use is a complex phenomenon with evident and different class-specific patterns, which highlights the value of explainable AI in providing detailed and interpretable insights for targeted intervention.

4.3.7 LIME Local Explanation :

The LIME explanation explains an individual prediction through localized representation of predicted class probabilities and feature-level contributions that affect the model decision. In the left panel it can be observed that, in this case, for curiosity is the main motivation (probability 0.52), followed by Th:Peer Pressure (p=0.20) and then Motivations:Other reasons (p=0.12).

In the middle panel, we explain that a subset of features influences the prediction in either

direction toward Easy Availability or Not Easy Availability. Indicators such as the Easy Availability (Yes, to some extent), Psychological Stress and Financial Pressure make a positive contribution for the Easy Availability class, however Family Relation and Employment related factors push lowers the prediction. This comparison demonstrates several factors are involved in the final decision.

The actual feature values of the sample individual are described in the right panel and demonstrate moderate financial pressure, psychological stress and presence of family problems, with partially free access. Such numbers are consistent with our analysis as they also support the contribution invariance between input factors and the model decisions. This Figure 4.3.7 LIME Local Explanation for Drug Use Prediciton:



Figure 4.3.7: LIME Local Explanation .

And factorsÂ such as Lack of Awareness, Lonely or Depressed, Special Prof/Trades or Gender performed weakest negatively; that is they wereÂ least important in this case. Overall, the LIME results provide some rationale why the model's decision leans one way or the other based on important factors in behavior (i.e., peer influence and curiosity) providing interpretable justifications for the individual-level prediction.

4.3.8 Permutation Importance :

Permutated importance - the most valuable features obtained by permuting importance method (ranked according to drop of F1-score after their random permutation). Larger weights correspond to a higher influence of the feature on the model's predictive performance. We find the Age and Start Age as the most sensitive features, permuting them leads to the highest drop in model performance. This emphasizes the importance of age-related factors and early-onset for predicting reasons for drug addiction. Family Relation and Peer Pressure (high) also become of significance, which highlights the influences from family relations and peers. Social and emotional factors, such as Love Failure, Cultural Pressure or Satisfaction, complement the model and demonstrate that addiction behavior is influenced by a combination of personal/,or emotional/environmental determinants. Compare this with attributes pertaining to companion type and location that have smaller effects (presumably secondary), as the calculated coefficients suggest. This Figure 4.3.8 demonstrate Permutation Feature Importance for Drug Use Prediction:

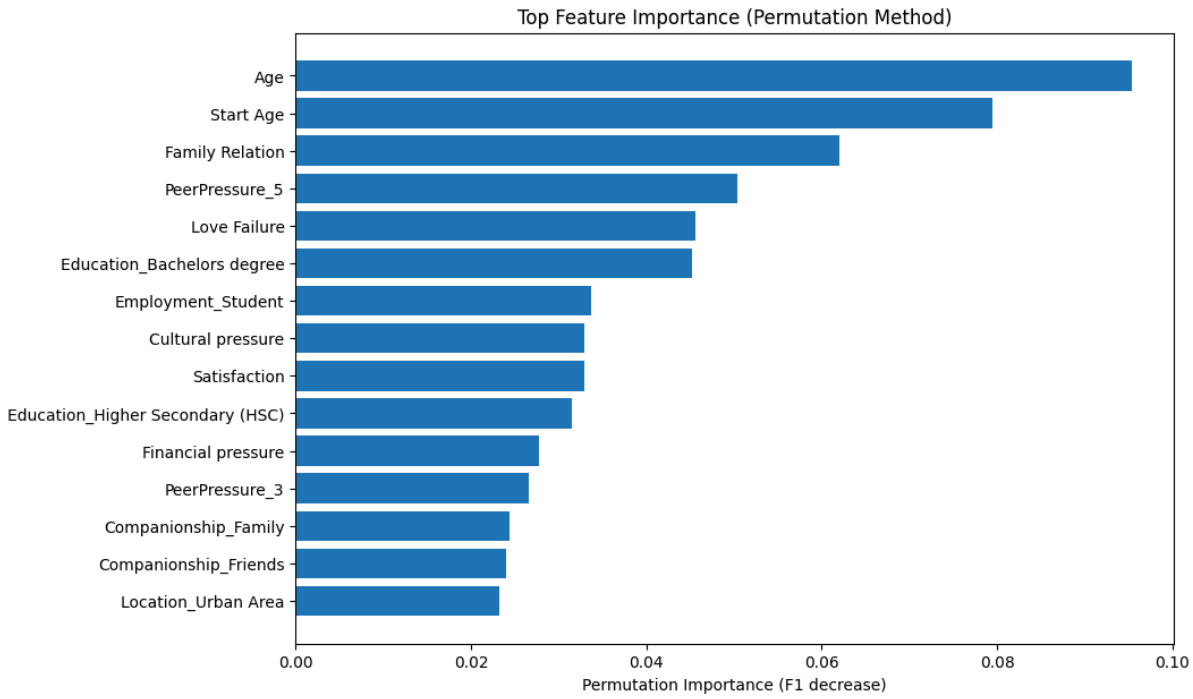


Figure 4.3.8: Permutation Feature Importance .

Finally, the permutation-based analysis demonstrates that the model mainly depends on age-related social and emotional features and it complements our SHAP based findings by performing a strong model-agnostic validation of feature importance.

4.3.9 Grouped SHAP Feature Importance :

The grouped SHAP is the sum of the features' SHAP importance (grouped into 4 higher level groups: SE [Socio-Economic], PS [Psycho-Social], Environment, and Others). The combined SHAP effect is the sum of contributions in features of each group for the model prediction.

According to the results, Socio-Economic factors are first then Psycho-Social in sequence of most affecting drug addicts predication. It means that economic status, academic and/or work situations, relationship with family members as well as psychological burden or emotion become key factors of determining addiction behaviors. Environmental determinants, including availability and location, demonstrate moderate but significant contribution whereas the Other category is a relatively less important determinant. This Figure 4.3.9 presentes Grouped SHAP Feature Importance for Drug Addiction Prediction:

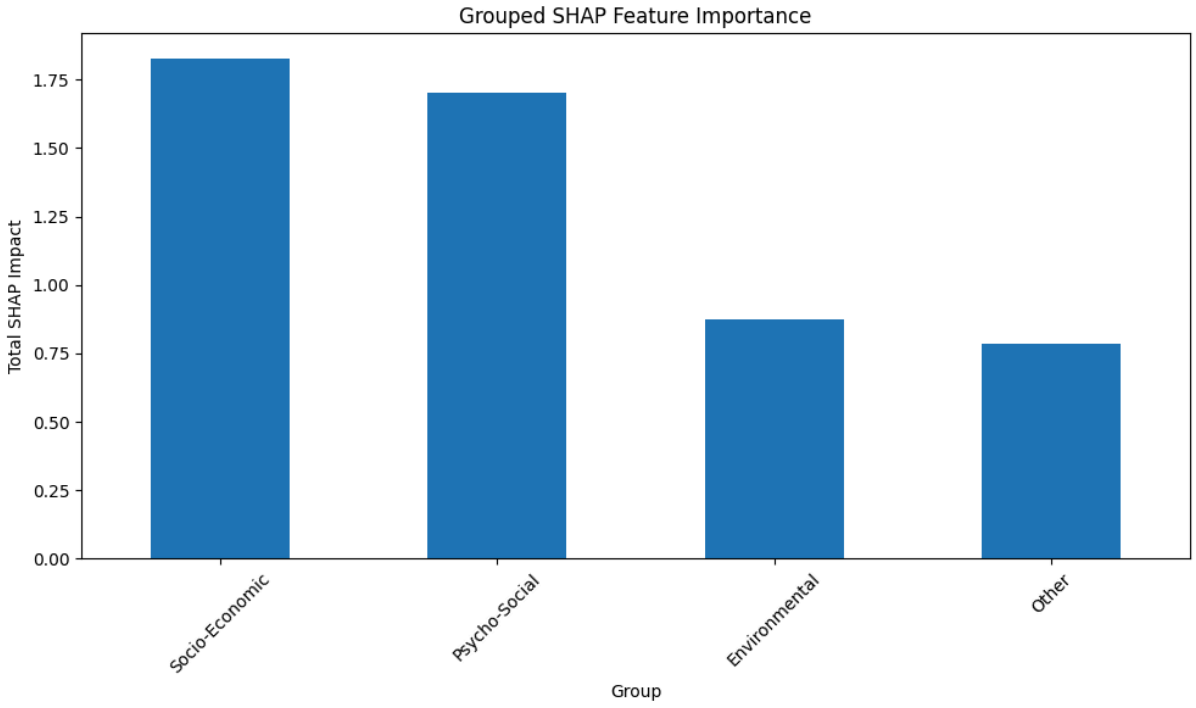


Figure 4.3.9: Grouped SHAP Feature Importance for Drug Addiction Prediction.

A combination of socio-economic and psycho-social factors, rather than just environmental ones characterises drug addiction generally according to this grouped analysis. By summarising SHAP values at the macroscopic level, the figure offers a policy-relevant view that can justify intervention targeting specific social, economic and psychological aspects.

4.4 Statistical Tests :

Statistical Validation of Features

Statistical significance of the relationships between features and target variable (the reason for drug addiction) were confirmed with multiple hypothesis testing by ANOVA, chi-square, Pearson correlation and Spearman correlation. The results are summarized below.

4.4.1 ANOVA Test Results (Numerical Features vs. Reason) :

The results of ANOVAs are informed of statistically significant differences among numerics, regardless that most have reached significance, such as numerical variables on age type, psychological and emotional reasons from socio-environmental influences. This indicates that the mean differences of these features are significantly different in terms of addiction cause. On the other hand, satisfaction is not significantly different -- there is little variation across classes. This Figure 4.3.1 displays ANOVA Test Results(Numerical Features vs Reason) for Feature Significance

:

Table 4.4.1: ANOVA Test Results (Numerical Features vs. Reason) .

Feature	F-Statistic	p-value	Significant (p < 0.05)
Age	2.6991	0.004763	Yes
Start Age	4.0159	0.000066	Yes
Family Relation	6.3060	0.000000	Yes
Financial Pressure	2.1842	0.022627	Yes
Psychological Stress	3.3893	0.000525	Yes
Life Failure	5.5939	0.000000	Yes
Love Failure	4.7486	0.000006	Yes
Cultural Pressure	3.5529	0.000307	Yes
Satisfaction	1.1591	0.320468	No

4.4.2 Chi-square Test Results (Categorical Features vs. Reason):

Significant associations were found between reasons of addiction and certain categorical variables (education, peer pressure, companionship, family problems and drug availability) as shown by Chi-square analysis. In contrast, gender and geography are not statistically associated with it, but they tend to explain addiction motivations more in social and environmental context other than personal or demographic characteristic. The Figure 4.4.2 below express Chi-square Test Results (CategoricalFeatures vs Reason) for the Feature Significance:

Table 4.4.2 : Chi-Square Test Results.

Feature	Chi-square	p-value	Significant (p < 0.05)
Gender	12.9892	0.792213	No
Employment	54.2948	0.025758	Yes
Education	444.5624	0.000000	Yes
Location	15.4711	0.078783	No
Companionship	172.9729	0.000000	Yes
Family Problem	63.8487	0.000000	Yes
Peer Pressure	182.7630	0.000000	Yes
Easy Availability	100.0914	0.000005	Yes
Category	1424.0000	0.000000	Yes

4.4.3 Pearson Correlation Results (Numerical Features vs. Reason Encoded)

The Pearson's correlation coefficients in numerical variables and the encoded drug addiction reason are weak linear relationship and non-significant ($p > 0.05$). The low correlation coefficients suggest that addiction reasons cannot be linearly related to individual numerical variables, which verifies the problem of linear-correlation analysis and invokes the necessity for using non-linear machine learning and explainable AI strategies. The Pearson Correlation Results for: this Figure 4.4.3 is illustrated in:

Table 4.4.3: Pearson Correlation Results.

Feature	Pearson r	p-value	Significant
Age	−0.0913	0.082948	No
Start Age	−0.0170	0.747023	No
Family Relation	0.0874	0.096717	No
Financial Pressure	0.0642	0.223129	No
Psychological Stress	−0.0803	0.127038	No
Life Failure	0.0326	0.536232	No
Love Failure	−0.0873	0.097323	No
Cultural Pressure	−0.0509	0.333729	No
Satisfaction	−0.0359	0.495359	No

4.4.4 Spearman Correlation Results

Weak, non-significant monotic relationships were seen between all numerical predictors and the encoded drug addiction: reason according to scores from Spearman rank correlation analyses ($p > 0.05$). The low Spearman ρ values indicate that the relationship of changes in numerical variables (age, start age, psychological stress, financial pressure and emotional factors) with decent trends is not strictly monotonic for addiction reason classes. These results shed light on the drawback of rank-based statistical techniques in modeling addiction reasons and emphasize that non-linear machine learning and explainable AI frameworks are much more capable of capturing the intricate underlying patterns. This Figure 4.4.4 illustrates Spearman Correlation Results :

Table 4.4.4: Spearman Correlation Results.

Feature	Spearman ρ	p-value	Significant
Age	−0.0918	0.081018	No
Start Age	0.0465	0.377290	No
Family Relation	0.0533	0.311462	No
Financial Pressure	0.0636	0.227024	No
Psychological Stress	−0.0767	0.145385	No
Life Failure	0.0692	0.188835	No
Love Failure	−0.0727	0.167633	No
Cultural Pressure	−0.0631	0.231228	No
Satisfaction	−0.0435	0.408814	No

Overall Discussion

Overall, the multi-variant statistical results indicate that both numerical and categorical features are significantly different across addiction reasons (as identified by ANOVA and Chi-square tests), but their relationship is largely non-linear or complex based on the weak correlation coefficients. These results strongly support the application of non-linear machine learning models and XAI strategies for identifying latent patterns that drive drug addict behavior.

4.4.5 Root Cause Extraction Table :

The SHAP-based analysis indicates that Start Age is the most important factor, thus highlighting that the age when people begin to use drugs significantly impacts their addiction behaviors. Age additionally contributes significantly, indicating that life stage is an important factor for addiction trajectories. Looking at the emotional factors, Life Failure and Love Failure seem to be strong contributing factors reflecting as personal losses and emotional trauma playing a major role in drug addiction risk. Figure 4.4.5 SHAP Analysis-based Detection of Root Causes of Drug Addiction:

Table 4.4.5: SHAP-Based Identification of Root Causes .

Rank	Feature	Mean SHAP Impact	Impact Direction
1	Start Age	–0.320952	Decreases Risk
2	Age	–0.213168	Decreases Risk
3	Life Failure	–0.164491	Decreases Risk
4	Love Failure	–0.145759	Decreases Risk
5	Cultural Pressure	–0.126733	Decreases Risk
6	Family Relation	–0.094716	Decreases Risk
7	Psychological Stress	–0.093863	Decreases Risk
8	Peer Pressure (Level 5)	–0.078441	Decreases Risk
9	Financial Pressure	–0.073146	Decreases Risk

These strongest are socio-cultural and relational factors (Cultural Pressure and Family Relation), indicating the importance of a cultural context as well as family relations. Psychological Stress and Financial Pressure demonstrate the impact of mental health and monetary strain, whereas High Level peer Pressure (Level 5) emphasizes the robustness of social influence. These values are consistently negative, which means that differences in these features have significant impact on the model's prediction direction; this is consistent with the fact that drug addiction is a complex interplay between age-related, emotional, psychological and socio-social factors.

4.5 Summary

We have shown an automated system to predict drug addiction with model evaluation across multiple machine learning models in this chapter. The models were validated with classical performance metrics, cross-validation and stability analysis, from which XGBoost was identified as the most robust. To interpret model prediction to predict major risk factors that was further validated through characterizing with permutation importance and statistical analysis, explainable AI approaches (SHAP and LIME) were used. Generally the proposed method demonstrates an open and honest way to unlock causes leading to drug addiction, helping with preventive intervention or guide by data accordingly.

Chapter 5

Engineering Standards and Design Challenges

This chapter summarizes the engineering standards guiding the project and the key design challenges that shaped our engineering decisions.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

When I started to use machine learning and XAI, I found that good practices when coding and using data in a standard way saves you plenty of time as much as unexpected disappointments! That's why I referred to a few industry standards:

1. **PEP8 (Python Coding Guideline):** I've written my code to follow PEP8 for readability and maintainability. This saved me from ugly scripts and it was way easier to debug.It also helped to structure the complete pipeline—preprocessing, training, evaluation and explainability.
2. **UTF-8 Encoding and CSV Format:** I had a corpus in Bangla and English. The data already would be broken, or display weird characters, without correct UTF-8 encoding. Because of that, CSV felt like the right format to use; It guaranteed I had valid data for an software to read.
3. **Scikit-learn Standard API:** Quick Start With scikit-learn's uniform approach (fit, predict, score) it was easy for me to make a pipeline. It also facilitated the integration with SHAP and LIME as these libraries assume that the model adheres to some conventions.
4. **IEEE Referencing Style:** Because this is a research article, I desired my writing to appear professional. IEEE is the one made me prevented my citations clean and believable.

5.1.2 Hardware Standards

(Note: I didn't use any fancy equipment for this project. I conducted most of my work on] my own laptop:

- Intel Core i5 processor
- 8GB RAM
- SSD storage

And when I needed more compute muscle, I turned to Google Colab which provides GPU access sometimes.

This configuration was sufficient for my dataset size and the models I trained. I didn't want to open a GPU server or use AWS/GCP, but they were overkill. So I have written a lot of this already My passing setup still does all the tests.

5.1.3 Communication Standards

Communication protocols were applied in the project to ensure that during each exchange among different components of any information over a medium, data was transferred and received without loss. Given that it was a private dataset about human actions, we had to monitor the information flow between all parties involved in the framework.

Secure Data Collection :

- Answers to this questionnaire were collected on the secure Hypertext Transfer Protocol Secure (HTTPS) Google Forms, enabling the answers to be transmitted in a secure manner and not accessible by unauthorised respondents.
- There were no personal identifications, maintaining full confidentiality and anonymity.

Structured Data Storage :

- All the data were saved in CSV/XLSX formats that make it sound and possible to keep everything tidy, and tidied up without any errors.
- That permitted an effortless access to machine learning tools without any distortion or loss of structure.

Controlled Data Sharing :

- The shared folder via Google Drive was the only place where any shared all research files/materials were stored (an encrypted, access controlled repository).
- This was put in place so that sensitive material could not be mistreated or lost, and so that only certain people had access to it.

Internal Communication Between Components :

- Pandas DataFrames and Python dictionaries were used to facilitate the easy flow of data between code modules during model development/preprocessing/explainability.
- This was a good deal because it helped to reduce the number of errors and was easier to debug.

Privacy-Focused Processing :

- All calculations (training of ML, XAI analysis and evaluation) in the local/secure (e.g., Google Colab) environment.
- All the data was kept only within the app for total confidentially and it was not sent to third party servers or any external AI tools for the same purpose.

Clear and Interpretable Outputs :

- Final outputs with SHAP and LIME explanations were laid out in a way that is easy to understand, even for non experts (e.g., rehabilitation workers or decision makers).
- It was a clear forward to the communication of learning and there were no misunderstandings.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The research sheds light on how and why people, particularly young adults, are susceptible to drug addiction. By employing the XAI methodologies (SHAP, LIME), in the research is clearly defined what socio-economic/psychological/environmental state mostly has an impact on addiction.

This has direct benefits:

- The flowchart may help families recognize early signals.
- Rehabilitation centers could personalize treatment based on individual risk factors.
- People may just become more sensitive to their own soft spots."
- Policy makers receive scientifically-based recommendations for prevention programmes.

5.2.2 Impact on Society & Environment

Societal Impact :

- Impact Statement This work has the potential to have measurable impact on public health and community health by:
- Reducing crime, violence and disruption linked to addiction.
- Aid government bodies such as DNC to develop focused interventions.
- Supporting resource prioritization among NGOs and mental health agencies.
- Strengthening the effectiveness of awareness programs with evidence-based data.

Environmental Impact :

- 100% digital workflow → less than no footprint; environmental win!
- No bodily waste, toxic chemicals or toxic materials employed.

5.2.3 Ethical Aspects

Drug dependency is a taboo subject. For this reason, rigorous ethical standards were pursued:

- Personal information (name, phone number or address) was not gathered.
- Participation in the survey was voluntary and anonymous.
- Data was never made public, nor stored in an insecure fashion.
- Finally, XAI offered fairness in the sense that biased or discriminatory behavior of any model could be avoided.
- Interpretability facilitated avoiding the misinterpretation of ML predictions.

5.2.4 Sustainability Plan

The project is designed to remain useful and expandable in the long term:

- The model can be retrained as new data becomes available.
- The pipeline is modular—new ML algorithms or XAI tools can be integrated easily.
- The low-cost computational setup makes it suitable for long-term use in academic and public health environments.
- The framework can be adapted for other issues such as alcoholism, smoking, or mental health.

5.3 Project Management and Financial Analysis

This work included multiple phases—data collection, interviews, statistical analysis, machine learning model construction, explanation of explanations (XAI), and raising awareness. These various elements had to be carefully and constantly managed in their financial aspects; require rationing of resources and justification for each cost. In this section, a cost breakdown of the project is displayed and alternative budget paths are proposed, with statements about financial decisions and long-term value system model.

Table 5.3. Expense Category and Cost Summary

Serial	Expense Category	Description	Estimated Cost (TK)
1	Data Collection	Travel to rehabilitation centers, coordination with DNC-authorized staff, phone calls, field communication, and follow-up interviews.	10000
2	Printing Costs	Printing research forms, questionnaires, consent papers, awareness posters, and final report copies.	3000
3	T-Shirt Costs	T-shirts used for awareness campaigns and team identification during field activities.	3000
5	Additional Field Interviews	Expanding dataset with more rehabilitation centers.	5000
		Total =	21000

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

This section relates the proposed research to classifications of complex problem-solving from the Washington Accord. Unveiling the underlying causes of drug addiction using Explainable AI In order to unravel the root causes of drug addiction by means of this XAI approach, sophisticated analytical techniques, and computational intelligence are exploited for combining tailored interdependencies between the interrelated sociological, economic and psychological factors, thus constituting a complex engineering challenge.

Table 5.1: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
✓	✗	✓	✓	✓	✓	✓

Mapping with Knowledge Profile

Properly identifying and predicting drug dependence needs to incorporate multidomain knowledge to fulfill the EP1. The avatars aims include statistical and mathematical analysis, machine learning expertise, socio-behavioral understanding, software pipeline implementation practice, practical skills in the real-world development of engineering pipelines extending biology to methodology. Accordingly, it is highly related with Knowledge Profile categories K2, K3, K4, K5, K6 and K8 in Table 5.2.

Table 5.2: Mapping with knowledge Profile.

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
✗	✓	✓	✓	✓	✓	✗	✓

5.4.2 Engineering Activities

The detection of the risk in drug addiction of an individual is a multi- faceted engineering challenge that requires sophisticated tools from computation and analytics, inter-disciplinary knowledge and profound social implications. The mapping of the problem to complex engineering works and the justification proposed for each category are provided in Table 5.3.

Mapping with Complex Engineering Activities

In this work, we aim to mount the proposed research of detection of causes for drugs addiction problems on the categories of Complex Engineering Activities (EA). The work demands new computational methodologies, efficient use of disparate computational sources and expertises, and the use of modern data-driven methods with societal consequences.

Table 5.3: Mapping with Complex Engineering Activities.

EA1 Range of re-sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓	✓	✓	✓

5.5 Summary

The standard load of drug addiction analysis and prediction system This chapter has introduced the engineering standards applied in construction of drug addiction analysis and prediction system, and determined that implementation of this design was justifiable. Social, ethical and environmental implications of the study were addressed, and potential contributions to evidence-based preventive and early intervention measures with a low ecological footprint. A cost-effectiveness analysis supported the financial viability of the project, as openly accessible data sets, open-source tools and low computational requirements were used.

The chapter also related the research to the complex engineering problems, knowledge profiles and engineering activities being addressed with an indication of its interdisciplinary character and analytical focus. In general, the chapter supports the feasibility, sustainability and technological adequacy of the proposed concept.

Chapter 6

Conclusion

This chapter concludes the study by summarizing the key findings derived from statistical analysis, machine learning, and Explainable AI techniques, and by highlighting the overall significance and implications of the research outcomes.

6.1 Summary

This research finds that interpretable machine learning can successfully uncover contributing factors of drug addiction, like behavior, psychology, social and environment. The proposed approach was based on a pipeline that included data pre-processing, feature encoding, high imbalance of classes control and training several models: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and XGBoost. The performance indicated that the ensemble and boosting models— in particular, XGBoost had the highest generalization capability for predicting addiction risk based on high F1-score, ROC–AUC as well as its stability across different random states.

Explainability was embedded using SHAP and LIME, which led to the interpretation of the most influencing factors for addiction predictions. Influential factors including peer pressure, curiosity, family problems, emotional stress and ignorance were found as the main factors. Such insights into explainability not only serve as a confirmation of the model's internal logic, but also promote transparency by lighting up explanations that policymakers, educators and people involved in health care can use to comprehend the drivers of addiction. It is of note that the integration of machine learning and XAI represents a potential promising strategy in early detection, intervention optimization and decision support for drug abuse prevention.

6.2 Limitation

While the presented model offers important insights into the aetiological pathways of drug dependence, it has several limitations: One, the dataset was relatively small for analysis and may not represent all addiction behavior. Some characteristics were based on self-report and may have been affected by recall bias, which could affect prediction accuracy. Furthermore, the use of single-source dataset (instead of aggregating two or more different sources from demographic and clinical datasets) might limit the generalizability of model on other targeted population.

The model relies on conventional machine learning methods without considering the temporal features or real-time behavior, which are useful for improving prediction performance. In addition, the XAI interpretations rely on completeness and quality of feature inputs; if relevant psychological or environmental variables are omitted, explanations may be partial. Lastly, this platform has not been tested in the real world or by expert clinical evaluation, which is a prerequisite for its use in rehabs and prevention programs.

6.3 Future Work

Future improvements can significantly enhance both the research value and practical utility of the proposed system. One major advancement would be developing a full-stack web or mobile application where the trained model can make real-time predictions for high-risk individuals. Integrating a secure backend database—such as MySQL, PostgreSQL, or MongoDB—would allow storage of user records, prediction histories, and role-based access for counselors, educators, or administrators. Deploying the entire system on cloud platforms like AWS, Azure, or Google Cloud would make the service widely accessible and scalable.

From a research perspective, future work may incorporate larger and more diverse datasets, including clinical assessments, socioeconomic indicators, family histories, and behavioral patterns collected via sensors or mobile applications. Advanced deep learning models or hybrid architectures could be explored to capture complex dependencies in addiction behavior. Additionally, interactive dashboards with enhanced SHAP/LIME visualizations can provide policymakers with better interpretability. Lastly, collaboration with psychologists, rehabilitation experts, and public health professionals is essential for validating the system in real-world settings and establishing it as a reliable decision-support tool for addiction prevention.

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Appendix

Approval Application Form for Rehabilitation Centers :



Daffodil International University
A landmark to create the Future
Faculty of Science and Information Technology

Date: 15/07/2025

To
The Chief Consultant
Central Drug Addiction Treatment Centre
Department of Narcotics Control
441, Tejgaon I/A, Dhaka

Subject: Request for Permission to Collect Research Data

Dear Sir,

With the respect and honor I, Md Shah Jalal, Senior Lecturer, Dept. of CSE, Daffodil International University (DIU), Dhaka, would like to mention that currently I am supervising a group of final-year research students Md. Abubakar Rahman and Shoaib Hossain, who are pursuing a research on **Explainable AI in drug addiction**.

As part of their undergraduate thesis, they are pursuing a research study on the patterns, causes, and predictive analysis of drug addiction in Bangladesh. To enrich the quality and authenticity of their academic work, it is essential that they collect real-world data from reputed rehabilitation and treatment institutions like the Central Drug Addiction Treatment Centre, which operates under the Department of Narcotics Control, Ministry of Home Affairs.

In this regard, I kindly request your permission to allow Mr. Rahman and Mr. Hossain to collect relevant, anonymized data (if possible) and observe or interact with the center under your guidance. Please be assured that all collected data will be used strictly for academic purposes and will be kept confidential, following standard ethical guidelines.

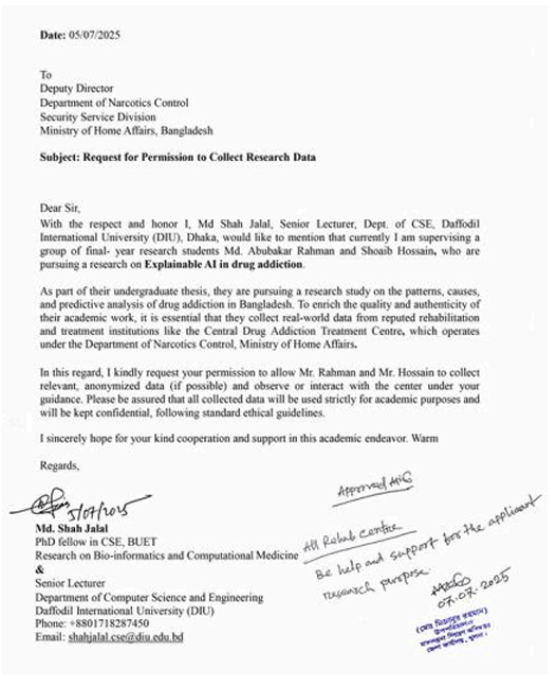
I sincerely hope for your kind cooperation and support in this academic endeavor.

Warm Regards,

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Date: 05/07/2025

To
Deputy Director
Department of Narcotics Control
Security Service Division
Ministry of Home Affairs, Bangladesh

Subject: Request for Permission to Collect Research Data

Dear Sir,
With the respect and honor I, Md Shah Jalal, Senior Lecturer, Dept. of CSE, Daffodil International University (DIU), Dhaka, would like to mention that currently I am supervising a group of final- year research students Md. Abubakar Rahman and Shoaib Hossain, who are pursuing a research on **Explainable AI in drug addiction**.

As part of their undergraduate thesis, they are pursuing a research study on the patterns, causes, and predictive analysis of drug addiction in Bangladesh. To enrich the quality and authenticity of their academic work, it is essential that they collect real-world data from reputed rehabilitation and treatment institutions like the Central Drug Addiction Treatment Centre, which operates under the Department of Narcotics Control, Ministry of Home Affairs.

In this regard, I kindly request your permission to allow Mr. Rahman and Mr. Hossain to collect relevant, anonymized data (if possible) and observe or interact with the center under your guidance. Please be assured that all collected data will be used strictly for academic purposes and will be kept confidential, following standard ethical guidelines.

I sincerely hope for your kind cooperation and support in this academic endeavor. Warm

Regards,

Md. Shah Jalal
PhD fellow in CSE, BUET
Research on Bio-informatics and Computational Medicine
&
Senior Lecturer
Department of Computer Science and Engineering
Daffodil International University (DIU)
Phone: +8801718287450
Email: shahjalal.cse@diu.edu.bd

Approved MIC
All Rehab centre
Be help and support for the applicant
research purpose.
07.08.2025



Daffodil International University
A landmark to create the Future
Faculty of Science and Information Technology

Date: 20/07/2025

To
The Authority
EMPATHY
Empathy Drug Addiction Treatment
& Mental Health Care Society
House-18, Road-6/A, Sector-05,
Uttara Model Town, Dhaka-1230, Bangladesh

Subject: Request for Permission to Collect Research Data

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Professor Dr. Shah Jalal Khan
Head
Department of CSE
Daffodil International University

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Date: 03/08/2025

To
The Authority
SOBER LIFE
Drug Addiction Treatment Support
And Rehabilitation Center
Section # 06, Block # D, Road # 12,
House # 03, Pallabi, Mirpur, Dhaka-1216

Subject: Request for Permission to Collect Research Data

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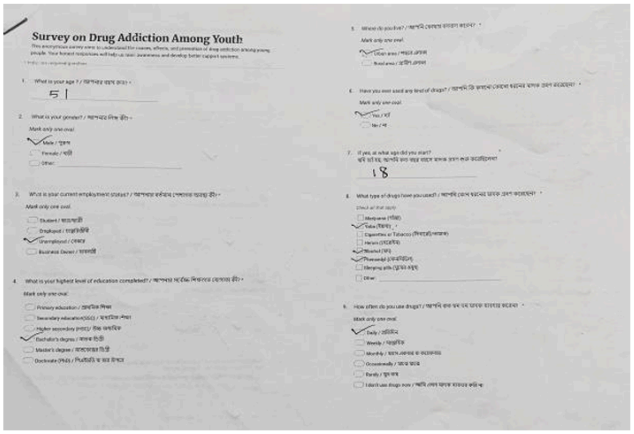
Warm Regards,

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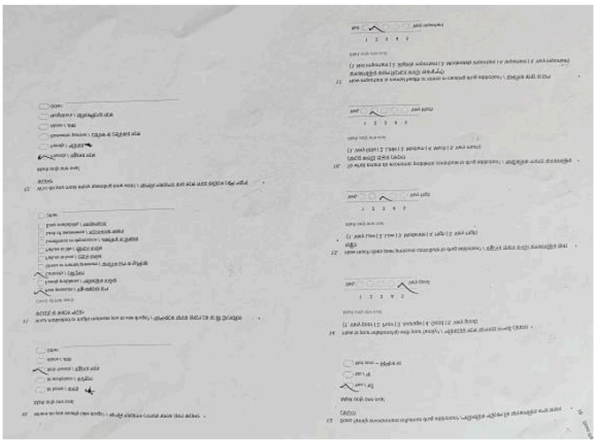
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Daffodil International University

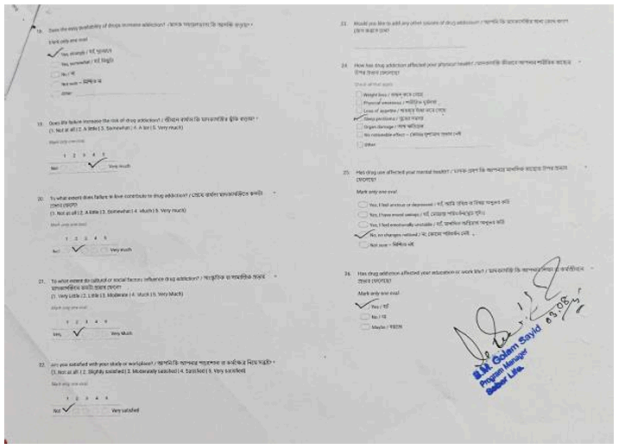
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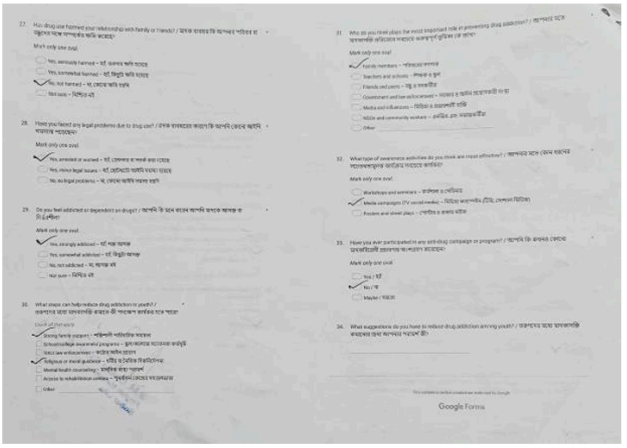
Data Collection Survey Form :



Survey on Drug Addiction Among Youth







Google Forms